Vitality as the Efficiency of Self-Paid Self-Modeling

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Abstract

Complexity is ubiquitous, the living rare: stars, hurricanes, crystals, and language models are richly structured, far from equilibrium, yet none pays, through its own dissipation, for its world-model. Closed *self-payment* separates the vital from the merely complex; here we propose its quantitative operationalization. Vitality requires two conditions: the predictive efficiency of self-modeling — the fraction of retained environmental memory still predicting the future — and self-payment, whereby holding obsolete memory carries a thermodynamic cost, so a corrupted model returns as the system's own decay. Requiring the model and its paying dissipation to share one physical boundary distinguishes the scale from established programs — assembly theory, integrated information, teleodynamics, the free energy principle — each taking one axis seriously, none demanding this identity of boundaries. A two-stage procedure — structural screening by a triad of invariants, then vitality verification — covers six paradigm cases from star to metropolis, with the biosphere as a control limit and a reproducible *Escherichia coli* chemotaxis computation. The picture is asymmetric: structural potential is nearly universal, closed self-payment loops rare. Consequences follow: extraterrestrial-life searches must detect such loops atop chemical biosignatures; artificial systems' ethical status turns on when one begins paying for its model; the complex–vital boundary falls at positive efficiency under a closed loop. Developed for the stationary regime, it generalizes to non-stationary dynamics in companion work (Andriishin 2026). It is open to refutation along several independent lines, including a predicted power-law dependence of evolutionary-adaptation rate on self-modeling efficiency.

Keywords: definition of life, vitality, self-payment, nostalgia, predictive information, thermodynamics of computation

1 Introduction

The concept of a "living organism" has, in contemporary biology, no generally accepted formal definition. Historically it rested on a cellular substrate — carbon chemistry, membrane organization, metabolic closure — but its boundaries are blurred both from below and from above. From below: viruses lack autonomous metabolism, yet they participate in evolutionary dynamics (Pradeu 2018); prions and replicator RNA systems reproduce without cellular infrastructure; protocellular and abiogenetic transitional forms make it impossible to draw a sharp boundary between chemistry and biology (Plante 2025). From above: biospheres, symbiotic communities, and anthropogenic megastructures display marks of collective auto-regulation and evolutionary adaptation without any single cellular referent (Mossio et al. 2016). In response to these difficulties, the philosophy of biology of the last two decades has developed a robust **epistemological pluralism**:

- registries of a hundred or more competing definitions of life with no signs of convergence (Trifonov 2011);
- the argument for the need of a "theory of life" instead of a dictionary definition (Cleland 2019);
- the anti-definitionist line of "life-as-historical-lineage": life as a single monophyletic branch from LUCA, rather than a list of marks (Mariscal and Doolittle 2020);
- the conception of the organism as a biologically individuated system (Bich and Green 2018; Pradeu 2018).

No definition gains unanimous acceptance; there is a growing demand for operational criteria capable of applying to nonstandard candidates — astrobiological findings, artificial systems of complex information processing traditionally called cognitive, and hybrid ecological complexes (Chirumbolo and Vella 2026).

The present work proposes for this role an operational measure — the dimensionless predictive efficiency of self-modeling $\eta_v = I_{\text{pred}}/I_{\text{mem}}$, and a vitality verdict as the pair (S, η_v) of the self-payment predicate S and positive efficiency $\eta_v > 0$. Before unfolding this construction (§ 2), we outline the landscape of programs against which it is built.

Vitality is formalized by a line of technical programs, each taking up one of its sides: the thermodynamic ("life feeds on negentropy", Schrödinger 1944), the Bayesian (FEP: Friston 2005, 2010), the combinatorial complexity of assembly (assembly theory: Marshall et al. 2017; Sharma et al. 2023), the causal closure of information (Φ: Tononi 2004; Tononi et al. 2016), the self-emergence of norms (teleodynamics: Deacon 2011), the scale of goal-like behavior (cognitive light cone: Levin 2019). These are not mutually exclusive hypotheses: each loses discriminating power at the boundaries — FEP applies to a household thermostat (Baltieri and Buckley 2019) and to a Watt governor (Bruineberg et al. 2022), while the narrow NASA definition (Joyce 1994; Cleland and Chyba 2002) is silent about substrate commonality (a detailed comparison — § 4). Hence the bridge to our measure: to join the thermodynamic payment and the quality of the retained model into a single dimensionless scale.

The parallel **origin-of-life** program — the hypercycle (Eigen 1971; Eigen and Schuster 1979), autocatalytic sets and RAF networks (Hordijk and Steel 2004; Hordijk

2019), dissipative adaptation (Perunov, Marsland, and England 2016), major transitions (Maynard Smith and Szathmáry 1995), geochemical scenarios (Pross 2012; Smith and Morowitz 2016; Kauffman 2019) — operationalizes the abiotic→biotic transition as a self-sustaining informational-energetic loop over a geochemical flux; η_v does the same in the thermodynamic register, where the positivity of η_v is a necessary condition for the moment of the origin of life (§ 4.8; § 6).

The same limits appear in astrobiology: biosignatures based on disequilibrium (O_2/CH_4), isotopes (δ^{13}C), and chirality have abiotic alternatives, and the debates over the arsenate claim for strain GFAJ-1 (*Halomonadaceae*) (Wolfe-Simon et al. 2011 [retracted by *Science* 2025]; Reaves et al. 2012), over methane on Mars (Webster et al. 2018; Korablev et al. 2019), and over phosphine on Venus (Greaves et al. 2021; Villanueva et al. 2021), absent a vitality criterion, reduce to the interpretation of particular proxies — just as the dispute over the vitality of language models does.

From another side, the same question is approached by AI alignment: under what architectural conditions an optimizing agent begins to optimize its own continuation (Bostrom 2014; Hubinger et al. 2019; Carlsmith 2022). The object is overlapping, the angles different: AI safety describes the behavioral marks of the loop, whereas η_v and the predicate S describe its thermodynamic co-characteristic (a structural parallel, not a causal-generative connection; § 3.4).

The present work defines vitality as the efficiency of retaining one's own model by one's own dissipation.

Self-modeling is used here in a sense distinct from the phenomenal self-model of Metzinger (Metzinger 2003): it denotes the structural condition under which the model of the environment, the dissipation that pays for it, and the carrier that suffers from its errors belong to one physical system — the "self" is the carrier of the model and of its price, not the content of the model. Likewise, *vitality* here is a strictly operational informational-thermodynamic property (the predictive efficiency of self-modeling η_v), not the *vis vitalis* of the vitalist tradition.

The central object is the dimensionless ratio

$$\eta_v = \frac{I_{\text{pred}}}{I_{\text{mem}}} \in [0,1],$$

where I_{pred} is the predictive information about the future of one's own environment, retained by the internal structure of the system, and I_{mem} is the total information of that same structure about the past and current trajectory of the environment (precise definitions and the link to the classical formulation of Bialek–Nemenman–Tishby (Bialek et al. 2001) — § 2.1). The ratio η_v is the fraction of retained memory that is still predictive; the complementary fraction $\nu = 1 - \eta_v$ is *informational nostalgia*, the ballast of outdated memory. The bound $\eta_v \in [0,1]$ for **passive** self-modeling is a consequence of the data processing inequality (DPI), not an energetic normalization and not a convention about scale (the active case is an open item, § 2.1). Thermodynamics enters as a **separate anchor**: by the thermodynamics of prediction (Still et al. 2012, § 2.2) retaining non-predictive memory commits the system to a dissipation of no less than $k_B T$ per such bit — here and only here does the Landauer scale $k_B T \ln 2$ appear, as the coefficient of a strict bound rather than as an exchange rate of energy for bits. Vitality is a pair of conditions: positive efficiency $\eta_v > 0$ **together with** a closed self-payment loop (§ 2.7).

The program unfolds in three constructive steps. The first is the operational definition of the vital scale as $\eta_v = I_{\mathrm{pred}}/I_{\mathrm{mem}}$ (the freshness of the retained model), explicitly distinguished both from indices of complexity (assembly index, Φ, I_v as a composite of three capacities) and from the self-payment predicate S: a positive η_v is possible even for a system without its own payment (as for a language model), but vitality requires the conjunction of $\eta_v > 0$ and $S = 1$. The second is the separation of two methodological levels: structural screening through the composite $I_v = \sqrt[3]{T_v C_v S_v}$ (the thermodynamic T_v, the informational C_v, and the topological S_v capacities; § 2.3) and vital verification through η_v (a computational economy, not a logical strengthening). The third is the explicit conventionality of the system boundary as a methodological instrument: η_v depends on the choice of boundary, and this dependence is productive — it allows one to distinguish "the vitality of an LLM as an agent" (equal to zero) from "the vitality of a corporation" (distinct from zero, but referring to a different object).

The novelty lies not in the separate components. Predictive information (Bialek et al. 2001) and the Landauer limit (Landauer 1961) have been known for more than half a century. Self-modeling goes back to Rosen's anticipatory systems (Rosen 1985); the requirement of one's own payment for the model converges with Taleb's skin-in-the-game (Taleb 2018) and Maturana–Varela autopoiesis (Maturana and Varela 1980). The novelty lies in the architecture of their joining along three points. The specific ratio $I_{\mathrm{pred}}/I_{\mathrm{mem}}$, to the best of our knowledge, has not been used as a measure of vitality in the Bialek–Nemenman–Tishby line. The requirement that the model and the dissipation that pays for it belong to one physical boundary (self-payment), with a closed error-return loop, in explicit form as a criterion of vitality, has, to the best of our knowledge, not previously been formalized — and neither Landauer nor Bialek worked with it. The interpretation of the resulting ratio as a vital scale is an independent step that none of the source frameworks took.

The nearest contemporary line — the thermodynamics of prediction and learning — needs to be set apart separately, since it operates with kindred quantities. Goldt and Seifert (Goldt and Seifert 2017) formalize the thermodynamics of learning a rule in a neural network, introducing a learning efficiency as the ratio of acquired information to its thermodynamic cost, bounded by unity; Takahashi and Hayashi (Takahashi and Hayashi 2026) develop a yield of acquiring structural information about the environment (η_E, bits-per-joule) and a requirement of boundary closure — payment for the low-entropy resource crossing the system boundary. Both works share with the present one a common foundation (Landauer 1961; Still et al. 2012; Goldt and Seifert 2017), and therefore the demarcation is conducted not on the foundation, but on subject, dimension, and interpretation, and the lines are complementary rather than competing. Dimension: their η_E is bits-per-joule about the *environment*, the efficiency of acquiring external structure; $\eta_v = I_{\mathrm{pred}}/I_{\mathrm{mem}}$ is dimensionless, that is: the fraction of the retained *self-model* still remaining predictive — a measure of retention and obsolescence (nostalgia), not of acquisition. The status of self-payment: their boundary closure is a convention of accounting in the choice of boundary, whereas the predicate S (the co-location of the model, of the dissipation that pays for it, and of the error-return loop; § 2.7) is a constitutive criterion of vitality with an independent thermodynamic support on the Still bound. Finally, the bound $\eta_v \in [0,1]$ follows from the data processing inequality, not from a stochastic-thermodynamic bits-per-joule balance, and the thermodynamics of learning offers no vital interpretation of the pair (S, η_v).

The proposed definition is not a metaphysical declaration. η_v is an operational representation of vitality as a mode of existence, not its definition in essence. The standard philosophical distinction between the referent of a theory (a property of the system) and an operationalization (a relation between the system and the measuring frame) is analogous to the distinction "temperature as a measure of thermal energy" versus "the logarithm of the expansion of mercury by 100 graduations", or "the strength of an earthquake" versus "the Richter scale". η_v is an operational measure, disciplined by a concrete definition.

The present work deliberately restricts itself to the **stationary regime**, where the assumption of finite memory and ergodicity of the environment stabilizes the volume of retained memory and makes informational nostalgia stationary. In this regime the formalism yields sharp demarcation answers and testable predictions. The nonstationary regime — growing memory, a drifting environment, a cumulative budget — lies beyond the scope of this work and is developed in the companion work (Andriishin 2026). There, Lemma 2 is proved on the inevitability of informational nostalgia in the OU class (Ornstein–Uhlenbeck) of slow drift under the technical conditions listed there. A conjecture is also formulated about the adiabatic asymptotics of $\dot{\eta}_v^{\text{excess}}$ for systems without periodic memory reset. The applicability of the apparatus of the present work is limited to systems with an ergodic environment and finite memory; for biospheres on eonic windows, AI systems learning under drift, and civilizations with a growing archive — consult the nonstationary extension.

The structure of the paper: § 2 unfolds the formal framework; § 3 applies the procedure to six paradigm-case systems (diagnostic test cases, each isolating one load-bearing condition); § 4 compares with neighboring programs (assembly theory, IIT, teleodynamics, FEP) and positions it relative to the lines of organisational closure, the origin of life, and contemporary demarcation debates; § 5 formulates the falsification conditions; § 6 discusses the consequences for astrobiology, AI ethics, and complex systems; § 7 sums up.

2 Theoretical Framework

2.1 The Central Quantity: Predictive Efficiency of Self-Modeling

A living system holds in its internal structure an implicit model of its environment. Part of what is memorized still predicts the environment's future — it is useful; the remainder is a trace of a past from which the environment has already moved on — ballast. Vitality is measured by **what fraction of the retained memory remains predictive**: the higher the fraction, the more "alive" the loop of self-modeling, whereas pure ballast without prediction is no longer a working model.

The formalization of this intuition rests on two information quantities. Let M_t be the configuration of those internal degrees of freedom of the system X at time t that pass counterfactual ablation (§ 2.2) as belonging to the modeling loop (a functional channel, not the full internal configuration — for a bacterium this is the methylation pattern of the Tar receptors, not all $\sim 10^{10}$ degrees of freedom of the cell), and let X_E be the signal of its environment. In the spirit of Still, Sivak, Bell, and Crooks (2012) we introduce

$$I_{\text{mem}}(t) := I\Big(M_t;\ X_E^{\le t}\Big), \qquad I_{\text{pred}}(t,\tau) := I\Big(M_t;\ X_E^{[t,\,t+\tau]}\Big). \tag{1a}$$

Here I_{mem} is **memory**: how much the internal state knows about the past and current trajectory of the environment (and this memory is rightly called a *model*, rather than a raw correlation, only when M_t is a minimally sufficient statistic of the past for predicting the future — the model-hood criterion: § 2.3); I_{pred} is **predictiveness**: what part of this knowledge pertains to the environment's future over the horizon τ. The narrow proxy operator M_t is a functional channel, not the full mutual information $I(X;X_E)$ over all degrees of freedom: the latter would be many orders of magnitude larger owing to **nonfunctional couplings** with the environment (heat exchange, mechanical contact, diffusion) that statistically bind X and X_E but carry no predictive information about its future behavior. The predictive component also differs from the classical predictive information of Bialek–Nemenman–Tishby (Bialek et al. 2001), defined as the mutual information between the past and the future of a single process, $I(X_E^{[t-\tau,\,t]};\, X_E^{[t,\,t+\tau]})$; the relation between them is given by the data processing inequality (see below) and is discussed further as a bridge to the Bialek framework.

The central quantity of the article — the **predictive efficiency of self-modeling** — is defined as the ratio of these two quantities:

$$\eta_v(t,\tau) := \frac{I_{\text{pred}}(t,\tau)}{I_{\text{mem}}(t)} = 1-\nu(t,\tau), \qquad \nu(t,\tau) := \frac{I_{\text{mem}}(t)-I_{\text{pred}}(t,\tau)}{I_{\text{mem}}(t)}, \tag{1}$$

where ν is **informational nostalgia**, the fraction of retained memory that has ceased to be predictive. *That is:* η_v is not "bits of prediction per joule," but the fraction of what the system remembers that actually pertains to the future; ν is its complementary fraction of pure ballast.

The bound on this quantity is a theorem, not a normalization convention. Memory is formed from the past with internal noise, $M_t = f(X_E^{\le t}, \zeta_v)$, where the noise carries no information about the environment's future beyond what is already contained in the past: $\zeta_v \perp X_E^{[t,\,t+\tau]} \mid X_E^{\le t}$ (it is precisely this conditional independence, and not the stronger requirement of "no feedback," that is the exact premise needed; disentangling these two conditions — supplementary § S2.1). Then the chain $M_t \to X_E^{\le t} \to X_E^{[t,\,t+\tau]}$ is Markov, and by the data processing inequality (DPI; information about a random variable does not increase under deterministic processing) $I_{\text{pred}} \le I_{\text{mem}}$. Hence

$$\eta_v \in [0,1], \qquad \nu \in [0,1] \qquad \text{— by construction (DPI).}$$

That is: the $[0,1]$ normalization is a consequence of an information inequality.

The bound $\eta_v \leq 1$ is derived for **passive** self-modeling, in which the Markov property of the chain $M_t \to X_E^{\leq t} \to X_E^{[t,\,t+\tau]}$ holds. The active case (freely-swimming *E. coli*; active-inference agents, § 7), in which M_t causally drives the environment's future through action and thereby breaks the DPI chain, remains **an open item**: conditioning on a fixed policy does not restore the Markov property (the action $a = \pi(M_t)$ makes M_t dependent on the environment's future), and the bound requires a transition to **directed** (transfer) information $T_{M \to X_E}$ — how much the model's own state M causally contributes to the environment's future beyond its past — a feedback generalization via the Sagawa–Ueda bound (2010); the setup — § 2.7 below.

Interventional (do) legitimacy of the passive measurement. The openness of the active case does not render the operational measurement of η_v meaningless: the immobilized protocol of § 3 fixes not a "simplified" but a strictly defined quantity, because it realizes a **do-intervention** in Pearl's sense (Pearl 2009). The exogenous clamping of the action $\mathrm{do}(a)$ — a controlled perturbation that fixes a by an external hand — severs the causal arrow $M_t \to a \to X_E$ and restores the Markov property $M_t \to X_E^{\leq t} \to X_E^{[t,\,t+\tau]}$; by the DPI

$$\eta_v^{\mathrm{do}} := \frac{I\Big(M_t;\ X_E^{[t,\,t+\tau]}\ \Big|\ \mathrm{do}(a)\Big)}{I_{\mathrm{mem}}} \in [0,1]. \tag{1b}$$

That is: under an intervention that clamps the action exogenously (and **not** conditioning on the observed policy — these are different operations: passive conditioning leaves the path $X_E^{\mathrm{past}} \to M_t \to a \to X_E^{\mathrm{fut}}$ open, whereas $\mathrm{do}(a)$ cuts the arrow entering a from M_t), the loop is passive again and the ratio lies correctly in $[0,1]$. The denominator meanwhile remains the observational I_{mem}, not an interventional one: the clamp $\mathrm{do}(a)$ in the present and future does not change the **already-formed** correlation of M_t with the environment's past (the past segment $X_E^{\leq t}$ on which I_{mem} is evaluated is formed **before** the clamp), so $I\Big(M_t;\ X_E^{\leq t}\ \Big|\ \mathrm{do}(a)\Big) = I_{\mathrm{mem}}$, and the DPI under intervention closes $\eta_v^{\mathrm{do}} \leq 1$ with numerator and denominator in **one** interventional regime (the immobilized protocol of § 3 fixes both). For *E. coli* the physical realization of $\mathrm{do}(a)$ is **immobilization**: a cell pinned to a substrate does not swim, its state does not move the ligand concentration, and the arrow $M \to a \to X_E$ is severed. The immobilized FRET protocol of Sourjik–Berg (§ 3) is therefore **not a simplification but an operational realization** of $\mathrm{do}(a)$, making η_v a proper $[0,1]$ quantity; a freely-swimming cell measures a different, active quantity, not bounded by unity. At the same time, η_v^{do} is **not a new theorem**, but a standard do-reduction of the active loop to a passive one, fixing only the precise sense of the legitimacy of the passive measurement. One must not conflate η_v^{do} with the thermodynamic efficiencies of the same feedback literature: a feedback efficiency bounded by unity does exist — the thermodynamic $\eta_X = \dot{I}/\dot{S}_r^X \leq 1$ (Horowitz and Esposito 2014, eq. (18)) and the sensory capacity $C = \dot{I}/T_{y \to x} \in [0,1]$ (Hartich, Barato, and Seifert 2016), with definitions in § 2.7 — but this is a **thermodynamic/sensory** quantity (information flow against entropy production, sensor precision), **not** the predictive freshness η_v; they cannot be identified.

Reconciliation of S and $\mathrm{do}(a)$. The clamp $\mathrm{do}(a)$, which suspends the **action branch** of the error-return loop (ii) for the duration of the η_v^{do} measurement, does not conflict with the assignment $S = 1$. The self-payment predicate S requires that the information retained by the model and the dissipation that pays for retaining it belong to one physical boundary of the system — self-feeding (condition (i)) and the return of model error into its own loop (condition (ii)); the full definition — § 2.7. This is a **dispositional** predicate of the *native* (freely-swimming) organism, established by a **separate** intervention — counterfactual ablation (§ 2.2) — and not by the do-clamp. The loop (ii) has **two branches** — the action channel (sensorimotor error return) and the dissipative Still branch; $\mathrm{do}(a)$ suspends **only the action branch** for the duration of the evaluation of η_v^{do} and does not cancel the dispositional S; that is, S and η_v^{do} assess one system by **different** protocols, and the apparent conflict "with a clamped, how is $S = 1$ asserted?" does not arise — the verdict $V(X) := S \wedge (\eta_v > 0)$ joins two different functions of the system, measured by two different interventions. For the immobilized preparation the **dissipative branch (ii) remains active** and holds $S = 1$: instantiated not through the severed action channel but through the **dissipative mechanism** of Still — distortion of the model $\to$ growth of the nonpredictive correlation $I_{\mathrm{nonpred}} \to$ a larger Still tax, paid by the cell's **own ATP** (condition (ii) of § 2.7 already admits "growth of obligatory dissipation **or** disintegration of X").

Before binding η_v to a fixed horizon, we introduce the system's characteristic window. The horizon $\tau_d := I_{\mathrm{pred}}/\dot{I}_{\mathrm{pred}}$ is the memory turnover time, the ratio of the stock of predictive information to the rate of its arrival; it is defined for **non-Markov** (memory-bearing) environments. For Markov environments $\dot{I}_{\mathrm{pred}} = 0$ (the horizon adds no predictive information beyond a single step), and the characteristic horizon is set by the environment correlation time τ_E. The empirical windows — the adaptation time of the signaling loop (seconds to minutes, within the life of a single cell) for a bacterium, a year for a city, the response time of a biogeochemical cycle (a year to a decade) for the biosphere — are methodological proxies for τ_d, whose consistency with the theoretical horizon is a separate calibration question (§ 3.5). Henceforth everywhere $\eta_v := \eta_v(t, \tau_d)$.

The quantity η_v has the status of a **within-class grade** of the freshness of the self-model — it measures how outdated the model of a given system is — **and not** of a cross-system absolute scale. The ratio $I_{\mathrm{pred}}/I_{\mathrm{mem}}$ penalizes the volume of memory, so it cannot be used to compare "a bacterium against a brain" directly: the cross-system demarcation of (synchronic) vitality is the work of the predicate S (§ 2.7): the binary test "vital or not" is performed by the predicate, while the numerical value of η_v within the positive region gives a comparative measure of the efficiency of model retention. Why self-payment specifically (rather than autopoiesis, RAF, or organisational closure) serves as the membership criterion, and why S is a deliberate ontological premise rather than an arbitrary definition, is examined in § 2.7 (epistemic status): information theory is responsible only for the **efficiency** of the model and does not itself determine **who pays** for its retention; the predicate S adds precisely this physical membership criterion, with a positive necessity argument (within-system culling of errors) and complementarity — not competition — with the line of organisational closure.

On the measurability of the denominator. The denominator $I_{\mathrm{mem}} = I(M_t; X_E^{\le t})$ is measurable from time series in the same way as the actual dissipation E_{actual} — a quantity accessible through calorimetry: for the chemotactic loop of *E. coli* in the passive (immobilized) FRET protocol of Sourjik–Berg it depends only on the covariances of methylation (a latent variable recovered from the FRET readout of kinase activity) and of the ligand (microfluidics); in the Gaussian approximation of the linear model it is estimated via the logdet of these covariances (the full proof-of-measurability and worked example — § 3).

The thermodynamic anchor (Still 2012). Memory ballast is not free: for a system to maintain a correlation with an already-irrelevant past, the thermostat must receive heat. Still proved the exact lower price of this ballast, and it serves in the present framework as a **separate thermodynamic anchor**, not part of definition (1). At the memory-update step $t \to t+1$, for the instantaneous quantities $I^{\mathrm{inst}}_{\mathrm{mem}} = I(M_t; X_{E,t})$ and $I^{\mathrm{inst}}_{\mathrm{pred}} = I(M_t; X_{E,t+1})$, Still proves an **equality** (his equation (14)), $\langle W^{t\to t+1}_{\mathrm{diss}} \rangle = k_B T\,(I^{\mathrm{inst}}_{\mathrm{mem}} - I^{\mathrm{inst}}_{\mathrm{pred}})$; for the full protocol (work and the subsequent relaxation) a nonnegative relaxation term turns it into a **lower bound** of Still — henceforth bound (2) (his equation (18)). The Still price is reckoned step by step — each step of memory update is paid for separately, and over a window these payments add up; therefore we distinguish the instantaneous nostalgia ν^{inst} (the payment of a single step) from the horizon nostalgia $\nu(t,\tau)$ (the fraction of ballast over the whole window):

$$\langle W_{\mathrm{diss}} \rangle \;\ge\; k_B T \left(I^{\mathrm{inst}}_{\mathrm{mem}} - I^{\mathrm{inst}}_{\mathrm{pred}}\right) \;=\; k_B T\, \nu^{\mathrm{inst}}\, I^{\mathrm{inst}}_{\mathrm{mem}}. \tag{2}$$

That is: each bit of instantaneous nostalgia costs the system no less than $k_B T \ln 2$ of dissipated heat — this is the strict thermodynamic meaning of retaining an aging model. Two caveats fix the precise status of the bound. First, (2) carries **instantaneous** nostalgia $\nu^{\mathrm{inst}} = 1 - I^{\mathrm{inst}}_{\mathrm{pred}}/I^{\mathrm{inst}}_{\mathrm{mem}}$ (the Still layer), not the horizon nostalgia $\nu(t,\tau)$ from (1); the cumulative dissipation over a window is the **sum** of the step-wise contributions, and not the product $k_B T\, \nu^{\mathrm{cum}} I^{\mathrm{cum}}_{\mathrm{mem}}$ (the step-wise structure of dissipation and its relation to the horizon $\nu(t,\tau)$ — supplementary § S2.1.1). Second, $k_B T$ enters here as the **coefficient of a bound from a theorem** — a multiplier in the inequality — and **not** as an exchange rate of energy for bits; the only legitimate appearance of the Landauer scale $k_B T \ln 2$ is the cost of erasing one bit inside (2). This anchor clarifies the connection of vitality with irreversibility — but in a subtle sense that is easily distorted by inversion. The living apparatus is selected **toward** the Landauer limit **stepwise** (Still 2012): biomolecular machines are near-reversible per step (on the order of $20\,k_B T$ per operation — Bennett 1982), and this stepwise efficiency is a **mark** of vitality, not its negation. What is excluded is only **global** reversibility of the whole operation: the load-bearing condition (ii) of the predicate S is the correction of model error, and correction (kinetic proofreading; Hopfield 1974; Ninio 1975) is thermodynamically necessarily dissipative (Bennett 1982; Landauer 1961: a perpetually-working useful machine cannot store its entire history), so a strictly reversible self-modeling loop **is unable** to correct error and therefore does not satisfy (ii) — both of its branches, "growth of obligatory dissipation" and "disintegration of X" (itself irreversible), trace back to one prohibition. That is, $S = 0$ here follows

by construction of the predicate, and not from a nulled magnitude of dissipation (the informational η_v meanwhile survives — the bound is from the DPI, not from erasure); in this limited sense vitality is plausibly **constitutively irreversible** — a thesis whose formal demonstration remains an open problem (the rigorous route and the stepwise/global distinction — supplementary § S2.1.1; the domain of definition of the T_v-anchor — § S5.1).

The connection of predictive information with dissipation runs through the **change in entropy and mutual information** (the Still bound (2); the generalized Landauer principle, Sagawa 2014, P03025), and not through a conversion of free energy into bits. In definition (1) there is no $k_B T \ln 2$ scale; the "Landauer" line appears legitimately — through bound (2), where and only where $k_B T$ enters. When $\dot{I}_{\mathrm{mem}} \to 0$ (memory is stabilized, $I_{\mathrm{mem}} \approx$ const) and under stationary drift the nostalgia ν reaches a constant level, while the cumulative dissipation grows linearly with a positive specific rate $\geq k_B T \varepsilon$ — the strict thermodynamic meaning of the ineliminability of drift in the stationary regime (supplementary § S2.1.1); the non-stationary regime is examined in the companion work (Andriishin 2026).

2.2 The System Boundary as a Methodological Convention

The scale η_v depends on the choice of the boundary of the measured system. This is a structural property, not a defect. For a large language model under the strict boundary "the weights at the moment of inference" (a forward pass without state retention) two conditions fail simultaneously: the own dissipation $E^{\mathrm{own}}_{\mathrm{actual}} \to 0$ (the power grid is external) and the channel of error return into the working loop is absent. The scale η_v under such a boundary is **undefined** through the simultaneous failure of both conditions, not "equal to zero through the numerator" (the formal count via the capacities T_v, C_v, S_v — § 2.3; the distinction between the strict and the operational reading of I_v^{int} versus I_v^{op} — § 2.5). Under a boundary that includes the data center and the corporation, $\eta_v > 0$, but the measured object is no longer the model.

The choice of boundary is methodologically disciplined. The boundary is drawn where the loop of decisions paid for by the system's own disintegration is closed. The conceptual predecessor is Pattee's *epistemic cut* and *semantic closure* (Pattee 1982, 2001): the boundary between the physico-dynamical level and the symbolic level, at which the system closes within itself the processing of its own symbols. A bacterial cell is a natural boundary, since distortion of its internal model of the gradient leads to the disintegration of precisely that cell. A city — because distortion of its model of flows returns as a crisis to precisely that administrative unit. The identity of the question determines the boundary; the choice of boundary sets what is measured.

The conventionality of the boundary is productive. The objection "with the right choice of boundary any system will turn out to be vital" is parried thus: the choice of boundary is the choice of whose vitality is being measured, and widening the boundary widens the responsible subject rather than dissolving the diagnosis.

The boundary and the Markov blanket. The Markov blanket of a system X (Friston 2019) is a statistically separating layer of sensory and active states. The framework's own construction of the self-payment boundary (§ 2.2) **does not inherit** the difficulties of the realist interpretation of the Markov blanket diagnosed in BDDB-2022

(Bruineberg et al. 2022) and Aguilera et al. (2022). The boundary in our framework is set by the **thermodynamic membership of the budget** E_{actual}, and not by a statistical partitioning of the distribution along a Pearl- or Friston-blanket; it is therefore orthogonal to the Pearl/Friston-blanket dilemma. Conflating the two notions leads to a category error in which informational separation is taken for energetic self-payment.

Counterfactual ablation. The discipline of the choice of boundary is operationalized by an interventional procedure in the spirit of Pearlian causality (terminologically — counterfactual ablation in the sense of mechanistic interpretability: remove a component, measure the effect; not to be confused with Reichenbachian screening-off). For a candidate c for inclusion in the boundary of X: c is subtracted from the physical realization; if the change in the rate of free-energy depletion $-dF_X/dt$ over a response time $< \tau_d$ is significant, c belongs to the boundary. The quantitative threshold: c belongs to the boundary when $|\Delta(-dF_X/dt)|/|-dF_X/dt|_{\text{base}} \geq \delta$, where $\delta \in [0.05, 0.1]$ is a methodological convention analogous to the significance threshold in regression analysis. For systems with smoothly distributed membership (parasite–symbiont–host, corporation–employees) the procedure returns a scale of degree of membership — the degree of membership $w_c = |\Delta(-dF_X/dt)|/\sum_i |\Delta(-dF_{X,i}/dt)| \in [0, 1]$.

An essential caveat about *what* the intervention measures: the effect of subtracting c is taken as the contribution to the **closed corrective regulation loop** — topologically, by whether c lies on the path "perturbation of the regulated variable $\to$ corrective action" with a response within time $< \tau_d$ — and **not** as a raw energy flow in itself. Without this refinement, a pure power source with the largest $|\Delta(-dF_X/dt)|$ (the thermostat's power grid, whose removal nulls all dissipation) would fall **inside** the boundary; the topological distinction of the loop leaves it outside — it *feeds* the dissipation but does not enter the loop responding to perturbation of the regulated variable (the bimetal and the relay enter, the power grid and the heater do not).

The procedure relies on a pre-theoretic candidate boundary — a minimal initial intuition (physical connectedness, the biological cell as a unit, an engineered assembly as a unit). Counterfactual subtraction refines the boundary by measuring the **relative** contribution of each component to the total dissipation of the connected system; convergence to a fixed point with respect to permutations of the subtracted elements yields a justified boundary. This is not a complete elimination of conventionality (for systems without a natural pre-theoretic candidate — a corporation, the biosphere — the procedure remains open), but an operational discipline of choice in most physical and biological cases.

Boundary membership $\neq$ self-payment. The two operations are separated and applied in strict order. The membership of a candidate c in the boundary is decided **solely** by c's contribution to the regulation loop ($|\Delta(-dF_X/dt)| \geq \delta$ under perturbation of the regulated variable — by breaking the loop upon subtraction of c, topologically, and not by raw energy flow) — prior to and independently of the question of the payer (i); the self-payment predicate S (§ 2.7) is assessed **afterward**, on the already-fixed boundary. The order "boundary $\to$ then S" breaks precisely the vicious circle "the boundary is chosen so that S comes out as desired": for the thermostat, ablation assigns the bimetal and the relay to the boundary (their subtraction breaks the regulation loop), and leaves the heater and the power grid outside — and this is decided without any appeal to self-payment; only **after** fixing the boundary does it emerge that $S = 0$, because the obligatory dissipation is borne by the grid, not the bimetal (full treatment — § 4.4). The residual conventionality (the choice of the candidate boundary, the threshold δ) is thereby retained (§ 2.7): the mechanism is disciplined, but not eliminated.

Application to the LLM — § 3.4 and Supplementary § S3.4; to the thermostat — § 4.4 and Supplementary § S4.4.

2.3 The Triad of Invariants

The positivity of η_v requires the simultaneous presence of three capacities.

Thermodynamic capacity T_v — the price of holding the model against erosion. Irreversible erasure of a bit physically costs no less than $k_B T \ln 2$ of free energy (Landauer 1961); the exergy flux is the payment against noise. The standard proxy is the specific exergy flux $P_D \cdot \eta_{\mathrm{ex}}$, normalized to mass or volume; in the tradition of Chaisson (Chaisson 2001) it is measurable from molecules to galaxies.

Informational capacity C_v — the quality of compression: the adequacy of the model to the environment and its updatability. The proxies divide into two classes. The first — entropy-rate proxies h_μ (normalized Lempel–Ziv complexity, Bandt–Pompe permutation entropy) — operationalizes the rate of environmental novelty, not predictive information. The second — proxies for I_{pred}:

- excess entropy $\mathbf{E} := \lim_{L\to\infty} I(X^{[t-L,\,t]};\, X^{[t,\,t+L]})$ — the asymptotic value of predictive information in the Bialek–Nemenman–Tishby formulation in the limit of infinite windows; on finite windows it is operationalized through block-entropy estimates $H(L)$ (Crutchfield and Feldman 2003);
- statistical complexity C_μ (Crutchfield's computational mechanics; Shalizi and Crutchfield 2001) — the entropy of the minimally sufficient statistic for prediction, satisfies $C_\mu \geq \mathbf{E}$ and serves as a complementary measure of structural memory;
- PCI (perturbational complexity index) of Casali (Casali et al. 2013) — an estimate of nonlinear integration on cortico-biological data. Adami's parallel program (Adami 2002) defines physical complexity as the mutual information between the genome and the environment — a biologically concrete special case of the "quality of the model of the environment," conceptually akin to C_v. The distinction of classes is essential for a consistent operationalization of η_v. In the stationary regime of the present work both classes serve as proxies for one C_v. The identity $\eta_v = C_v^{\mathrm{predictive}}/C_v^{\mathrm{static}}$ is correct only within the **second** class (computational mechanics), where both quantities are bits of retained structure: C_v^{static} is identified with the statistical complexity C_μ (structural memory, a proxy for I_{mem}), and $C_v^{\mathrm{predictive}}$ with the excess entropy $\mathbf{E}$ (the predictive part, a proxy for I_{pred}). The theorem $C_\mu \geq \mathbf{E}$ then yields $\eta_v = \mathbf{E}/C_\mu \in [0,1]$ consistently with the DPI. The proxies of the **first** class (h_μ, Lempel–Ziv) measure the rate of environmental novelty — this is a **separate** index, not the denominator of η_v. Thereby the central quantity of § 2.1 acquires a triadic form: the predictive efficiency is the fraction of retained structural memory (C_μ), remaining predictive ($\mathbf{E}$) — $\eta_v = I_{\mathrm{pred}}/I_{\mathrm{mem}} = 1 - \nu = C_v^{\mathrm{predictive}}/C_v^{\mathrm{static}}$; the identity is exact when M_t is a causally-sufficient

(minimally sufficient) representation, and is a proxy-approximation for an arbitrary functional channel M_t. It is precisely this caveat that carries the **model-hood criterion**, separating a *model* from raw memory: what is retained is a model of the environment, not an arbitrary correlation, exactly when M_t is a causally-sufficient (minimally sufficient) statistic of the past for predicting the future, that is, when I_{mem} is identifiable with C_μ (the entropy of the causal states, Shalizi and Crutchfield 2001), rather than with the raw $I(X; X_E)$ inflated by nonfunctional couplings (§ 2.1). Then $\eta_v = \mathbf{E}/C_\mu$ measures exactly that fraction of the minimally-sufficient memory which carries predictive structure ($\mathbf{E}$); this is the basis for calling I_{mem} a **model**, rather than mere memory — and its legitimacy is the more complete the closer the functional channel M_t is to the minimally sufficient statistic. In the non-stationary extension (Andriishin, companion work) the divergence of C_v^{static} and $C_v^{\text{predictive}}$ serves as a diagnostic signal of informational nostalgia (the full relation $C_v^{\text{predictive}} = (1-\nu)\, C_v^{\text{static}}$ — § 6).

Topological capacity S_v — the architecture of the connectivity of the error-return loop. The standard proxies are Newman modularity Q (Newman and Girvan 2004) with the indicative threshold $Q > 0.3$ as a conditional benchmark, requiring calibration by a null model (see below on the resolution limit, rather than a hard criterion), scale-freeness $\gamma \in [2, 3]$ for the degree distribution (Barabási and Albert 1999; Newman 2010 review; a critique of its universality in Broido and Clauset 2019), the hierarchical clustering coefficient $C(k) \sim k^{-1}$ (Ravasz and Barabási 2003). The algebraic connectivity λ_2 of Fiedler (1973), indexing the global diffusive connectivity, is a *secondary* proxy with an inverse interpretation: for modular architectures λ_2 is low (Mohar 1991), so it is used as an indicator of structural bottlenecks between modules, not as a measure co-directed with Q. An additional class of proxy — structural controllability (Liu, Slotine, and Barabási 2011) — is an independent-of-modularity slice of the topology of the error-return loop.

The efficiency of a hierarchically-modular architecture for retaining predictive information over a structured environment is formalized through the minimum description length (Mengistu et al. 2016): the hierarchy of topology compresses the description of the adaptive policy by an amount proportional to the depth of the environment's hierarchy. The hierarchical structure is validated through the predictive power of the missing-links decomposition (Clauset, Moore, and Newman 2008), which empirically links the S_v-proxies with the C_v-criteria and prevents their degeneration into a single measure. The empirical correlate is the neural connectome (Bassett and Sporns 2017; Lynn and Bassett 2019), demonstrating hierarchically-modular organization as a selective correlate of computational efficiency. Two extreme cases null S_v constructively (§ 5, Block 1 "Direct Refutations," item 3). The Erdős–Rényi random graph — the absence of a modular decomposition; redundant short links do not yield compression of the adaptive policy, despite the small-world diameter (Watts and Strogatz 1998). The regular lattice — the absence of cross-scale connection, a loss on large-scale propagation of error correction.

The empirical status of scale-free as a universal pattern is contested: on a corpus of ~1000 empirical networks a strict power law with $\gamma \in (2, 3)$, tested by the statistical methodology of Clauset et al. (2009), is confirmed in a minority of cases (Broido and

Clauset 2019; Holme 2019). The power-law distribution is one of the regimes of the required architecture, not the only one; the more general condition of the absence of a characteristic scale and the presence of a modular-hierarchical organization is what is relevant. The Q metric of Newman depends on the algorithm (greedy Newman (2004), Louvain (Blondel et al. 2008), Leiden (Traag et al. 2019)); the resolution limit problem (Fortunato and Barthélemy 2007) requires calibration of the threshold through a null model. In the present work Q is computed by the Leiden algorithm; the numerical calculation for the chemotactic network of *E. coli* — supplementary § S2.4 ($Q \sim 0.5$). The evolutionary origin of hierarchical modularity (introduced above through $C(k) \sim k^{-1}$ of Ravasz–Barabási) is selection pressure for the minimization of connection length (Mengistu et al. 2016) and environmental variability (Kashtan and Alon 2005).

The triad is not assembled from independent disciplines — it is forced by the nature of holding compression over time. Nulling any capacity nulls vitality on any scale: without payment there is no stake — the system's own price for the quality of the retained model, which makes its errors significant to the system itself (skin-in-the-game, § 2.7) — without quality there is no model, without architecture there is no return loop.

2.4 Normalization of Capacities by a Reference Cohort

The proxies in direct form do not lie on $[0, 1]$ and are not substitutable into the composite I_v without normalization. The specific exergy flux for T_v has the dimension $\mathrm{W}\cdot\mathrm{kg}^{-1}$ and is unbounded above. Excess entropy and C_μ for C_v are bits, unbounded above. Newman modularity for S_v lies in $[-1/2, 1]$; the Ravasz–Barabási hierarchicalness is a power-law exponent. λ_2 for a connected graph $\in (0, n]$ (for a disconnected one $\lambda_2 = 0$); as a secondary proxy it is not used in the makeup of the S_v-normalization. Without explicit normalization the formula (2b) and the caveat "$T_v, C_v, S_v \in [0, 1]$" are formally incorrect.

In the present work, percentile normalization by a reference cohort is adopted. For each capacity a reference cohort of systems of the same structural class is fixed; the normalized quantity is the percentile in the cohort's distribution:

$$\tilde{T}_v(X) = F^{\text{cohort}}_{T_v}\big(T_v(X)\big) \in [0, 1], \tag{2a}$$

where $F^{\text{cohort}}_{T_v}$ is the empirical distribution function of the proxy over the cohort; similarly for $\tilde{C}_v$, $\tilde{S}_v$. The zeroth percentile is the weakest representative, the unit percentile the strongest. This is a standard device of social indicators: the Human Development Index (United Nations Development Programme 2022) normalizes its components in the same way. The concrete cohorts for the six paradigm cases and the handling of edge cases (the biosphere without a cohort; the crystal and the hurricane with a constructive zero) — Supplementary § S2.4.

The percentile procedure carries three limitations. First: the choice of cohort is a convention. At the same time, the robustness of the difference by sign between two systems of one cohort does not depend on the choice of cohort given sufficient cohort breadth; it is precisely this robustness by sign — the substantive claim of § 3, not the concrete numbers. Second: the procedure loses information about the absolute values of the proxies and works only for comparison within a class. Across classes the numerical I_v are directly incomparable. Third: the procedure is inapplicable to systems without a natural cohort, where normalization by a theoretical upper limit is required. The sigmoid alternative $\tilde{T}_v = T_v/(T_v + T_v^{\mathrm{ref}})$ — Supplementary § S2.4.

Henceforth in the text T_v, C_v, S_v denote the normalized quantities $\tilde{T}_v, \tilde{C}_v, \tilde{S}_v$, unless explicitly stated otherwise; the tilde is dropped for readability.

2.5 The Composite I_v as an Index of Structural Readiness

The geometric mean of the three capacities

$$I_v = \sqrt[3]{\tilde{T}_v \cdot \tilde{C}_v \cdot \tilde{S}_v} \in [0, 1] \tag{2b}$$

is defined as the structural composite. Geometric averaging is chosen for three reasons: it remains in $[0, 1]$; it is symmetric under permutations; it vanishes when any one capacity vanishes (the dropping out of a single capacity collapses vitality entirely). Additive averaging does not preserve this property — a sum would mask the failure. The weights $w_T = w_C = w_S = 1$ are a convention; for unequal weights $I_v^{(w)} = (\tilde{T}_v^{w_T} \cdot \tilde{C}_v^{w_C} \cdot \tilde{S}_v^{w_S})^{1/(w_T+w_C+w_S)}$. A sensitivity analysis with respect to the weights is not carried out, since I_v is a methodological index of structural readiness for vitality, not a standalone vitality scale: the binary "alive or not" test is performed by η_v, not I_v. A positive I_v is a necessary condition for a positive η_v; it is not a sufficient one. Through standard proxies (Lempel–Ziv, exergy, network metrics), I_v yields nonzero values for any structurally complex dissipative system — the Sun, a hurricane, a black hole, an LLM. In its positional place, I_v coincides with the assembly index, Deacon's morphodynamics, and Tononi's Φ: a composite index of complexity, necessary but not sufficient.

An objection to the status of I_v might be that a composite of three necessary conditions would more naturally be interpreted as a standalone scale. However, through any reasonable set of industrial proxies I_v yields stably positive values for systems that are unmistakably non-vital. For example, the I_v^{op} of the Sun is positive through any standard proxy. The normalized Lempel–Ziv complexity of the GOES X-flux over the 11-year cycle gives ~ 0.5– 0.7 of the upper percentile of the cohort of main-sequence stars; the hierarchicality of the active-region network from SDO/HMI (Gheibi et al. 2017) is ~ 0.4. The specific number depends on the dataset and cohort, but stably $\in (0.3,\ 0.7)$. Hence I_v remains in the role of a *methodological* index; formally this is secured by a distinction between two ways of reading it: I_v^{op} (operational reading) — through standard proxies on observational data — is nonzero for any complex dissipative system. I_v^{int} (interpretive / strict reading) — with a strict check of T_v as own dissipation, C_v as a model of the environment in the technical sense, S_v as the architecture of a closed loop — is zero for any non-vital system by construction. A divergence of the two readings for a particular system is a diagnostic signal: the system is structurally complex but not vital, and the divergence points to the failing capacity.

2.6 The Two-Stage Procedure

Vitality is theoretically established by a single test — for the positivity of η_v. The two-stage procedure is a methodological convenience, not a logical necessity:

1. *Screening by I_v.* Cheap, through standard operational proxies; it culls structurally unsuitable systems (the crystal — vanishing of T_v; thermal equilibrium — vanishing of all three capacities). A computational filter, not a separate test.
2. *Verification of η_v.* Expensive, requiring identification of the boundary and of the system's own self-payment loop; applied to candidates that have passed the screening. The only logically substantive test.

The binary nature of the sign of η_v does not make the numerical value decorative. The objection "if the sign already delivers the demarcation, the number is redundant — the counterfactual ablation of § 2.2 establishes the sign without a formula" is parried by pointing to content of the magnitude that is irreducible to the sign. First, η_v within the positive region is a comparative metric of the efficiency of the self-payment loop: the order of magnitude of η_v for two systems of the same class carries content beyond the shared sign. Second, the magnitude enters the falsifiable quantitative prediction $r_{\text{adapt}} = A \cdot \eta_v^{\alpha}$ (§ 5), which the sign does not generate: the sign fixes neither the slope α nor the testable monotone dependence of the adaptation rate. The sign of η_v separates only the class without a model ($\eta_v = 0$); the binary demarcation of (synchronic) vitality among model-bearing systems is carried by the predicate S (§ 2.7), while the magnitude of η_v within $S = 1$ is an intra-class metric and a risky prediction; the latter is not derivable from the sign.

The structure of outcomes: passed both thresholds — vital (the bacterium, a forest, the brain, the biosphere, a city); passed the screening, failed the verification — structurally rich but non-vital (the Sun, a hurricane, a black hole; the LLM under the strict boundary "model as agent" — failure with $S = 0$); failed the screening — structurally unsuitable (the crystal, thermal equilibrium).

Refutation of any one capacity collapses both scales simultaneously: η_v through I_{mem} or I_{pred}, the predicate S through failure of T_v/the error-return loop, I_v through the geometric mean. Refutations of the two scales are one and the same diagnostic at two levels of resolution, not independent pieces of evidence.

The two-stage architecture carries a methodological risk. The class "structurally rich but non-vital" — the Sun, a hurricane, a crystal, an LLM, a black hole, the Belousov–Zhabotinsky reaction — risks turning into a Lakatosian protective belt: any refuting observation is absorbed by inclusion into the class. The program fixes a minimal discipline: every vanishing of η_v is accompanied by an indication of the specific capacity that failed. If the class expands without a structural explanation of each new inclusion, the program shifts from a progressive to a degenerating regime.

A possible counter-argument: the asymmetry of the two levels is an artifact of the soft proxies of I_v (Lempel–Ziv, permutation entropy). Were C_v operationalized more strictly — for example, through Tononi's Φ with a filter on predictive power relative to the system's own environment — the Sun would yield $C_v \to 0$. The answer: the matter is not the choice of one proxy but the class of available industrial metrics. All the proxies of network science, information theory, and non-equilibrium thermodynamics developed over the last forty years, when applied to a complex dissipative system, yield nonzero values. A proxy that unmistakably zeroes out the C_v of the Sun requires post-hoc calibration on the vitality hypothesis itself — a circle.

2.7 Self-Payment and Epistemic Status

The magnitude η_v tells how fresh the model is, but does not answer the program's central question — **who pays** for its retention. The bimetallic strip in a thermostat also "models" temperature, but the dissipation of maintenance is paid by the power grid; *E. coli* extracts its ATP itself. Vitality arises when a system **itself pays** the mandatory — per Still — price of retaining its own model. This is a thermodynamic form of organisational closure (Rosen 1991; § 4.7), in which the closure is realized by the physical loop model $\leftrightarrow$ dissipation itself, and not by a category-theoretic abstraction.

Formally, bound (2) makes the retention of non-equilibrium predictive memory dissipatively **mandatory**: $\langle W_{\text{diss}}\rangle \geq k_B T\, I_{\text{nonpred}}$, where $I_{\text{nonpred}} = I^{\text{inst}}_{\text{mem}} - I^{\text{inst}}_{\text{pred}}$. We introduce the **self-payment predicate** $S(X) \in \{0,1\}$, carrying error-return closure (the error-return loop, not merely "feeding"). It is essential that the boundary of the system X is fixed **independently of the question of the payer (i)** — by the counterfactual ablation of § 2.2 through the contribution to the regulation loop (a topological discrimination of the loop, not raw energy flux) — and is not chosen so as "to make S come out as desired"; this disciplines the circle "boundary $\leftrightarrow$ self-payment", though it does not eliminate it entirely — the choice of the candidate boundary and the criterion of membership in the regulation loop remain conventional (§ 2.2), and this conventionality is productive: the choice of boundary is the choice of whose vitality is being measured. At the fixed boundary, $S(X) = 1$ if and only if **both** conditions hold:

- **(i)** the mandatory dissipation of bound (2) is fueled by free energy that X **itself** extracts within its boundary (there is no external payer);
- **(ii)** distortion of the retained model **returns** as a rise of this mandatory dissipation or as the disintegration of X — a closed **error-return loop** (model distortion $\to$ bad action $\to$ loss of free energy by the system itself), operationalized by the counterfactual ablation of § 2.2.

Condition (ii) is load-bearing: it cuts off dissipative structures that merely *feed* but are not *corrected* (the flame, the Belousov–Zhabotinsky reaction, simple autocatalysis). Vitality is defined as a **pair**:

$$V(X) \;:=\; S(X) \;\wedge\; \big(\eta_v(X) > 0\big),$$

that is, the system **retains a predictive self-model** ($\eta_v > 0$) **and pays for its maintenance itself** ($S = 1$). *That is:* η_v answers the question "how good is the model", S answers the question "does the system pay for it itself"; the living is both. Here η_v **grades** vitality among vital systems (in the non-stationary case — through the dynamics of nostalgia $\nu(t)$, companion work Andriishin 2026), while S carries the **demarcation** of (synchronic) vitality (the full one — synchronic plus diachronic — being the "alive/life" convention of § 2.7, § 4.8).

The logical status of the conjunction. The conjunct $\eta_v > 0$ carries an independent demarcation only for systems **without a model** (the structural zero of the numerator:

the Sun and the crystal are cut off before the check of S). Among systems **with** a model, a closed error-return loop (ii) already presupposes nonzero predictive content, so that $S = 1$ practically entails $\eta_v > 0$; hence here the demarcation of (synchronic) vitality is carried by S, while η_v functions as a **grade**, not as a second independent condition. The conjunction is nonetheless not empty: $\eta_v = 0$ cuts off a class (systems without a model) inaccessible to the predicate S (a system may fail **both** axes at once — the structural zero of the numerator *and* the absence of self-payment: thus the virion-in-isolation carries $\eta_v = 0$ and $S = 0$ as a double failure, stronger than the single-axis cases). Therefore, in the demarcation table below, η_v for systems with $S = 0$ **and a retained model** is marked "undefined" — as a vitality grade it is meaningful only after passing the $S = 1$ screening, whereas as the formal ratio $I_{\text{pred}}/I_{\text{mem}}$ it is computable everywhere $I_{\text{mem}} > 0$.

It is precisely the self-payment condition that distinguishes the present framework from Still's. In Still's, the predictor and the dissipator may be **different** circuits (the demon and the environment being separate subsystems); here S **requires their coincidence** — the dissipation mandatory per Still must be borne by the same physical boundary X that bears the model. This "beyond-Still" requirement (co-location of model $\leftrightarrow$ dissipator) is the program's original contribution: the co-location of model $\leftrightarrow$ dissipator is expressed by an **explicit predicate on the dissipation of bound (2)** with the error-return loop (ii).

Active thermodynamic accounting of the S-side: a borrowed bound, not a discriminator of self-payment. The Still anchor (bound (2)) is derived for passive memory updating; in the active (feedback) form, where M acts causally on the environment, the active **accounting of work and dissipation** is carried by the strict inequalities of the information thermodynamics of bipartite systems — a borrowed apparatus, not a contribution of the present work (and, as clarified below, one that by itself does **not** discriminate self-payment). The local second law for the modeling circuit M (Horowitz and Esposito 2014, eq. (11)),

$$\dot{S}_i^M = d_t S_M + \dot{S}_r^M - \dot{I}^M \geq 0,$$

that is: the entropy production of the circuit M itself is non-negative, but its entropy balance is modified by the **information flow** $\dot{I}^M$ — the time derivative of the **symmetric** mutual information $I(M; X_E)$ (Horowitz and Esposito 2014), a term that redistributes entropy between M and the environment. This HE flow $\dot{I}^M$ **should not be identified** with the directed transfer-bound $T_{M \to X_E}$ below (Hartich–Barato–Seifert): these are two **different** borrowed objects — the symmetric flow in the HE entropy balance versus the directed transfer-entropy HBS on extractable work, not synonyms. Reading $\dot{I}^M$ as the active form of the "Still tax on I_{nonpred}" (the non-predictive correlation now being paid for not by the impersonal non-negativity of entropy, but by the term $\dot{I}^M$) is an **interpretive heuristic** of the present framework, not a result of HE (as is, below, the identification $I \leftrightarrow I_{\text{pred}}$ of Sagawa–Ueda, strictly correct only in the bipartite limit of directed information). The work extractable on the predictive side is bounded not by raw mutual information but by the **directed** transfer-entropy (Hartich, Barato, and Seifert 2014, eq. (37)),

$$-\sigma_M \le T_{M\to X_E},$$

that is: here σ_M is the entropy production of the **thermal reservoir** of the circuit M (the heat dissipated into that reservoir; $-\sigma_M$ is the work extracted from it, divided by T; in the stationary regime, $d_t S_M = 0$, it coincides with the full subsystem production of HBS, for which $\sigma^M = \dot{S}_r^M + d_t S_M$), and **not** the entropy of the environment X_E; and this work extracted by the dynamics of M does not exceed the flow $T_{M\to X_E}$ that the own state of M causally contributes to the future of the environment beyond its past. It is precisely the **actuator** direction $M \to X_E$ that is taken, since the matter concerns the feedback regime: here M causally **drives** the future of the environment, whereas the demonic extractable work in the purely **sensory** regime (the environment drives M) is conventionally bounded by the reverse flow $T_{X_E\to M}$, consistent with the sensory capacity $C = \dot{I}/T_{y\to x}$ above; for the active self-paying loop the former is relevant. The full feedback floor on dissipation is given by the generalized second law $\langle W\rangle \ge \Delta F - k_B T\, I$ (Sagawa and Ueda 2010, eq. (3)). The conclusion is more honestly formulated thus: in the transition to the active case the **pair framework** (S, η_v) remains meaningful, but its load-bearing parts change status rather than being "preserved" intact. Feedback thermodynamics (Horowitz–Esposito; Hartich–Barato–Seifert; Sagawa–Ueda) keeps the active **accounting of work and dissipation** determinate — it gives a bound on the work extractable by the self-paying circuit in the feedback regime as well — however, these inequalities hold for **any** bipartite system (a thermostat with $S = 0$ satisfies HE/HBS/SU exactly as *E. coli* with $S = 1$ does) and therefore **do not discriminate self-payment**: by themselves they do not distinguish $S = 1$ from $S = 0$. The demarcation is still carried by the predicate S — the presence of a payer (i) and of the error-return loop (ii), fixed by the counterfactual ablation (§ 2.2) — and **not** by these borrowed bounds. The magnitude η_v thereby **retreats** to the passively-defined η_v^{do} (under the intervention do(a), § 2.1); in the full active form its $[0,1]$ bound is **not guaranteed** (supplementary § S2.1.1). It is therefore correct to speak not of a "strict support" but of an active thermodynamic **accounting** (a bound on work/dissipation), borrowed and by itself not discriminating self-payment; the novelty of the program remains in the ratio $I_{\mathrm{pred}}/I_{\mathrm{mem}}$ itself and in the predicate of co-location of model $\leftrightarrow$ dissipator, not in this thermo-apparatus (Horowitz–Esposito; Hartich–Barato–Seifert).

The demarcation table:

System	η_v	S	Verdict
Thermostat / bimetallic strip	undefined ($S = 0$)	0 (the grid pays)	not vital
LLM (weights at inference)	undefined ($S = 0$)	0 (external compute)	not vital
Sun / flame / hurricane	$= 0$ (no I_{pred})	—	not vital — a stake without a model
BZ reaction / simple autocatalysis	undefined ($S = 0$)	(i) yes, **(ii) no**	not vital — *feeds* but is not *corrected*
Virion (outside the host)	$= 0$ (no I_{pred} / no autonomous model)	0 (does not pay itself)	not vital; virus + host — a reassignment of the subject
E. coli	> 0 (η_v^{do}, § 2.1)	1 (its own ATP)	vital
Biosphere / city / corporation	> 0 (magnitude undefined)	1 (self-funded)	vital (the grade requires MI-estimation — of mutual information)

When $I_{\text{mem}} = 0$, the convention $\eta_v := 0$ is adopted — no model, no vitality; the diagnosis of the Sun is a **structural zero of the numerator**, not a 0/0 indeterminacy. For grey zones (a cyborg, a managed ecosystem, a partially subsidized firm) self-payment can be graded by the share of own maintenance $\chi = E^{\text{own}}_{\text{maint}}/E^{\text{total}}_{\text{maint}} \in [0, 1]$ (energy for energy, without a bit $\leftrightarrow$ joule conversion), with a threshold $S = \nVdash[\chi > \chi_0]$; the primary object, however, is the binary S, while χ is its extension.

For active agents the predicate S additionally accounts for the energy of **action** (the sensorimotor loop), and bound (2) is taken in feedback form (Sagawa–Ueda 2010): for the active loop I_{pred} is what the agent can *cash out* into work (the Sagawa–Ueda gain is bounded by $k_B T\, I$ in terms of the measurement information I; the identification of I with I_{pred} is a heuristic, strictly correct only in the bipartite limit of directed information), while I_{nonpred} is the pure nostalgia tax (Still); the error-return loop (ii) is here naturally read as "model distortion $\to$ bad action $\to$ loss of the agent's free energy". The full formulation of the active case is a direction of future work (the feedback generalization, Sagawa–Ueda 2010).

Epistemic status of the section's claims. The claims of the work differ in status: theoretical deduction from established principles, empirical verifiability, speculative extrapolation with explicit falsification conditions. The claim about the domain of definition $\eta_v \in [0, 1]$ (§ 2.1) is a **deduction from the data processing inequality** (DPI) for passive self-modeling; the bound $\eta_v \leq 1$ is an **unconditional theorem** of an information inequality, not a conditional consequence of Landauer's principle. The geometric averaging of capacities (§ 2.4) is a formal choice based on symmetry and vanishing. The paradigm-case values of η_v for the bacterium, the biosphere, and astrobiological candidates are order-of-magnitude theoretical estimates with an explicit dependence on the window and the proxy. The city case (§ 3.6) is a paradigm-case order-of-magnitude estimate of η_v; the structural urban metrics in the spirit of

Bettencourt et al. (2007) and the empirical calibration of I_v for specific cities are the subject of separate work and are not asserted here. The speculative consequences — the search for cosmological vitality, the moment of vitality of an artificial system — are discussed in §§ 5–6 with explicitly formulated falsification conditions.

The load-bearing **self-payment postulate** — the model and the dissipation that pays for it belong to one physical boundary X (the predicate S, conditions (i)+(ii) above) — stands apart from the statuses listed: it is not a deduction from FEP or from Landauer's principle, but an explicit ontological premise of the framework, which neither Landauer nor Bialek made. Its epistemic status is a constructive choice; but this choice is not arbitrary and does not rest on demarcation force alone (cf. § 4.4: self-payment offers an alternative, thermodynamic ontology that does not inherit the difficulties of the Pearl/Friston-blanket construction diagnosed by BDDB-2022). Behind it stands a **positive necessity argument**. A retained model culls its own errors only where the stake on its quality — skin-in-the-game (Taleb 2018; § 1) — lies **inside** the same boundary that bears the model. If the price of retention is paid by an external circuit ($S = 0$), distortion of the model does not return as selective pressure on the system: the error-return loop (ii) is open, and the correction, if it happens at all, is performed by an external agent, not by X itself. Therefore self-payment is a **necessary** condition for the errors of the retained model to be culled inside the system itself. It is precisely this that separates the three classes: "feeds-and-is-corrected" (the living), "feeds-but-is-not-corrected" (the flame, the BZ reaction — (ii) open) and "models-but-another-pays" (the thermostat, the LLM — (i) open). The pair of conditions is thereby **minimal**: amputating (ii) leaves a dissipative structure without correction, amputating (i) — an externally-paid model; both yield the non-vital, so both conditions are necessary and together minimally sufficient for the thermodynamic side of vitality. The postulate asserts nothing stronger: $V(X)$ remains a discriminator of (synchronic) vitality, compatible with anti-definitionism (§ 4.9), not a definition of life.

This same argument sets the **relation to the organisational closure line** (autopoiesis, Maturana–Varela 1980; M,R-systems, Rosen 1991; constraint closure, Mossio–Montévil–Longo 2016; RAF, Hordijk–Steel 2004; § 4.7): all of them point to a **qualitative** closure — the loop produces and maintains its own components or constraints — without fixing its measurable thermodynamic condition. Self-payment does not compete with them and does not introduce "yet another definition": it is a **complementary measurable thermodynamic cross-section** of the same closure — a **necessary** thermodynamic condition, fixed in the physical loop model $\leftrightarrow$ dissipation itself (the Still bound (2); the active accounting above) — that is, a translation of their qualitative circularity into the operational predicate S with an independently fixed boundary (§ 2.2).

This postulate is the **only** load-bearing ontological premise of the (S, η_v) demarcation: the bound $\eta_v \leq 1$ follows from the DPI unconditionally (§ 2.1), without any additional premise. The independent thermodynamic mandatoriness of retaining ballast, in turn, is secured by the Still bound (2) — a separate anchor, not a postulate.

The falsification conditions of § 5 may be realized at two levels. Methodological: the current operationalization is untenable, a different proxy, window, or mutual-information estimation procedure is required. Ontological: the concept of vitality through the efficiency of self-modeling is untenable in substance. The program takes the position of moderate operationalism: any observation refuting η_v through the failure of a particular operationalization is first interpreted methodologically and is translated into the ontological only after the alternative operationalizations have been exhausted.

The asymmetry of falsification. The program is falsified through false-negative cases (a biological system conventionally recognized as alive for which $\eta_v = 0$ under all sanctioned operationalizations — § 5, Block 2, item 3) and through structural failures of the progressive predictions (§ 5, Block 3, items 1–2). False-positive cases (a system non-vital by conventional criteria with a stable $\eta_v > 0$) are formally excluded by construction: the counterfactual ablation of § 2.2, for systems without a closed error-return loop, returns C_v^{int} as undefined, and η_v is undefined. This asymmetry is a deliberately accepted price, narrowing the falsification surface of the program to two lines (false-negative penetration § 5 Block 2 item 3, and failure of the progressive predictions § 5 Block 3 item 1): η_v is a criterion of closure of the self-payment loop, and systems without such a loop receive no positive score **by construction**. If neither of the two lines turns out to be empirically realizable, the scale risks reducing to a conventional redescription — this risk is acknowledged explicitly (cf. § 2.6), not masked. An alternative criterion — for example, informational complexity per assembly theory or Φ — may assign a nonzero value to a non-vital system as well, which reflects the different content of these programs (structural complexity versus closure of the thermodynamic loop). Hence refutation of the η_v-program is possible only through **symmetric penetration** — the discovery of a system, conventionally alive, for which the self-payment loop does not close (§ 5, Block 2, item 3), or through failure of the progressive predictions on calibrated biological pairs (§ 5, Block 3, item 1).

Heredity and major transitions: what η_v does not close. The η_v scale formalizes the **synchronic** side of vitality — the retention of one's own model over the window τ_d. This is a necessary condition for every moment of the life of a single organism. The **diachronic** side — heritability and evolvability in the sense of major transitions (Maynard Smith and Szathmáry 1995) — lies beyond the present stationary model. The stationary regime fixes finite memory and ergodicity of the environment (§ 2.1), which exclude population-evolutionary dynamics with drift and the accumulation of hereditary information. The positivity of η_v is a necessary but not sufficient condition for a full biological "definition of life" in the sense of the NASA definition (Joyce 1994). A sterile worker bee or a mule have $\eta_v > 0$ (alive moment-to-moment) but are excluded from Darwinian dynamics at the population level. The scale is consistent with the biological convention about their being alive and does not violate it. A full definition of life requires $\eta_v > 0$ **plus** Darwinian dynamics at the population level; these are two distinct scales — the individual synchronic and the population diachronic — and both are necessary. The connection with the companion work (Andriishin 2026): the non-stationary extension provides the thermodynamic apparatus on which the major-transitions program (Maynard Smith and Szathmáry 1995) can be formulated as an open research problem (cf. § 4.8) — the joining of the two scales is not proved either in the present work or in (Andriishin 2026), but in their combination it acquires a formal basis.

3 Demarcation Tests

The two-stage procedure — structural screening by the triad of invariants (§ 2.3–2.6) and vitality verification by the pair (S, η_v) (§ 2.7) — is applied below to six paradigm cases ranging from a star to a metropolis, with the biosphere as a control limit. Each paradigm case is chosen so as to fail exactly one of the load-bearing conditions and thereby show **which** one. The load-bearing **positive** paradigm case is biological: the bacterium *E. coli* (§ 3.5), the only example with a computed estimate of η_v and a verdict of $S = 1$; the astrophysical and AI paradigm cases (§ 3.1–3.4) serve as **negative controls**, each isolating which of the two conditions of the pair (S, η_v) fails. The vitality verdict is formulated as a pair: $S(X) \in \{0, 1\}$ answers the question "does the system pay for the retention of its own model itself, and does the model's error return through its own dissipation" (§ 2.7, conditions (i)+(ii)), while $\eta_v(X) \in [0, 1]$ answers the question "what fraction of the retained memory is still predictive." Vitality is the conjunction $S \wedge (\eta_v > 0)$. The demarcation is carried by the **closure of the self-payment loop** S, not by the absolute smallness of efficiency. Taken as the predictive fraction $\eta_v = I_{\text{pred}}/I_{\text{mem}}$, the magnitude of η_v for most paradigm cases is quantitatively **undetermined** and requires MI-estimation of I_{pred} and I_{mem} of the corresponding loop; a computed illustrative estimate is given only by the bacterial example (§ 3.5). The smallness of η_v is by itself not a discriminating feature — and does **not** mean that η_v is large: the ballast fraction may be large, but this does not fix the magnitude of the predictive fraction. What distinguishes the vital from the non-vital is precisely the predicate S. A comparative program on the boundary of non-living/living collectives through informational architecture is developed in *Theory in Biosciences* (Kim et al. 2021); the six paradigm cases of the present section continue that line, adding to the informational axis of Kim–Valentini–Hanson–Walker the thermodynamic requirement of self-payment.

3.1 The Sun, the Neutron Star, and the Black Hole

The Sun is rich in structure — convective cells, magnetic loops, differential rotation — and passes the structural screening on the T_v side (dissipation $\sim 4 \cdot 10^{26}$ W) with room to spare. Vitality verification, however, zeros it out through η_v: the Sun has **no model of the environment** in the technical sense of § 2.1 — there is no functional loop M_t whose state would carry mutual information about the future of the environmental signal. At $I_{\text{mem}} = 0$, the convention of § 2.7 is adopted, $\eta_v := 0$: no memory of the environment — hence no predictive fraction of it either. Diagnosis: a structurally rich, non-vital system. The same applies to the neutron star, the interstellar medium, and the supermassive black hole at the center of the Milky Way. Here $\eta_v = 0$ is a **structural zero of the numerator** ($I_{\text{pred}} = 0$ in the absence of I_{mem}), not a $0/0$ indeterminacy — this is important to distinguish from cases where it is not η_v that is zeroed but the predicate S (the LLM, the thermostat).

The black hole yields the same negative verdict for a deeper reason: the very carrier of a model is absent. The stationary classical horizon is described by the no-hair theorem (canonical formulations 1967–1975; the key work Israel 1967): three parameters (M, J, Q) fully specify the external metric, leaving no room for I_{pred} about the environment. Quantum and non-stationary corrections — thermal radiation and the information paradox it engenders (Hawking 1975, 1976), soft hair (Hawking et al. 2016) — reveal a richer informational structure of the horizon, but do not restore a model of the environment in the sense of § 2.1: the corresponding degrees of freedom do not retain I_{pred} about the environment with a return of the damage through their own dissipation. The status of the unitarity of evaporation remains open (the information paradox: Hawking 1976 proves a *violation* of predictability, while modern resolutions — the Page curve, the islands program — restore unitarity), but for demarcation it is inconsequential: a model of the environment in the sense of § 2.1 is absent independently of its resolution. The accretion disk around Sgr A* ($M \sim 4 \cdot 10^6 M_\odot$) yields the same diagnosis: the environmental variables do not return as damage through a causal channel to the disk plasma. For all these systems $\eta_v = 0$ structurally (no I_{mem}, no I_{pred}); the question of S does not even arise — there is nothing to self-pay for.

3.2 The Hurricane, the Flame, and Autocatalytic Oscillators

A hurricane and a candle flame dissipate free energy — $\sim 10^{15}$ W for an average hurricane, $\sim 10^2$ W for a candle flame — and pass the T_v screening. But they have no internal model of the environment paid for by this dissipation: the dissipation of exergy is not accompanied by retention of I_{pred}, so $\eta_v = 0$ through the numerator ($I_{\text{mem}} = 0$, no functional loop M_t). The diagnosis is a **stake without a model**: the system pays but does not model. In this, the flame and the hurricane are the mirror case of the LLM, for which the ratio is the reverse (§ 3.4): there is a model there, but no own payment.

The Belousov–Zhabotinsky reaction and simple autocatalysis are a subtler case, and it is precisely on it that the load-bearing **condition (ii)** of the predicate S operates (§ 2.7). The BZ reaction oscillates stably and for a long time, feeding on supplied free energy — that is, it formally satisfies condition (i) of self-payment. But its oscillations **do not form a correction loop**: there is no channel in which a distortion of the retained pattern would return as a growth of obligatory dissipation or as a breakdown of the system. The error-return loop (ii) is not closed. The diagnosis is $S = 0$ through the failure of condition (ii): the system **feeds but does not correct itself**, repetition without correction. This separates dissipative structures that merely maintain themselves from those that retain and repair a model of the environment. (Epistemically: it is premature to speak of the magnitude of η_v here — there is no substantive loop M_t for which I_{pred} would be defined; the verdict is carried by $S = 0$.)

3.3 The Crystal and Thermal Equilibrium

A crystal retains its structure with a relative strain $\lesssim 10^{-6}$ over times $\sim 10^9$ years — but does not hold a **model of the environment**. Its internal configuration is homomorphic to its own lattice, not to the future of the environmental signal: $I_{\mathrm{mem}} \approx 0$ in the sense of § 2.1 (mutual information between the state and the environment's trajectory, not between the state and itself). And the crystal does not actively pay for the retention of this structure: in the strict thermodynamic sense it is close to equilibrium relative to the lattice, the current dissipation of retention ≈ 0. Thus the crystal fails both axes at once — there is no model of the environment ($\eta_v := 0$ at $I_{\mathrm{mem}} = 0$) **and** there is no own payment for its retention ($S = 0$, there is nothing to feed). Diagnosis: structure without a model and without payment.

Systems in full thermal equilibrium (an isolated gas, a closed black cavity) zero out everything simultaneously — the canonical negative control point of the screening.

Cairns-Smith and the crystal-as-replicator. The Cairns-Smith program (1982) regarded clay crystals as pre-RNA replicators: heritable lattice defects are passed to daughter crystals upon growth and splitting. Upon widening the boundary to "crystal + solution phase," the counterfactual ablation of § 2.2 for (crystal + ionic solution) yields $T_v > 0$ — the ion flux pays for the replication of defects. However, the error-return loop remains open: an error in the replication of defects **does not return** as physical damage to the lattice itself; there is no channel in which a distortion of the defect pattern would worsen the thermodynamic conditions of the crystal's retention. This is a failure of condition (ii) — $S = 0$. The diagnosis is a re-addressing of the object to (crystal + solution), analogously to the LLM-corp (§ 3.4), not vitality of the crystal-as-object.

3.4 The Large Language Model

The LLM is the strongest pure case of the new demarcation: a system with a **strong predictive model and** $S = 0$. A trained model is predictively strong — it compresses the regularities of its training distribution and predicts the next token with a quality far surpassing chance (large I_{pred}); the exact fraction $\eta_v = I_{\mathrm{pred}}/I_{\mathrm{mem}}$ of the retained representation, however, is not measured and is not asserted here. What is essential for the demarcation is not the value of η_v but the fact that the model exists and is predictive, whereas the loop of its payment is open. This is exactly why the LLM is the mirror of the hurricane: there, payment without a model; here, a model without own payment.

The decisive factor is the predicate S. Under the strict boundary "the weights at the moment of inference" (a forward pass without saving state), the obligatory dissipation of retaining and updating the model is paid for by the **external** power grid, not by the system itself: condition (i) of self-payment is violated. At the same time, there is no physical channel of error return into the working loop — the output tokens do not return as physical damage to the weights, the model's error does not deplete the carrier of I_{pred}: condition (ii) is violated as well. The result: $S = 0$ with the predictive model intact. The diagnosis is a **strong predictive model with an open payment loop**. This is a categorically different negative verdict than that of the Sun: there the numerator is zeroed ($\eta_v = 0$, structurally no model), here it is the predicate of self-payment, with the model intact and predictively strong.

Widening the boundary and re-addressing the object. Upon widening to "model + data center + corporation," the predicate S may become positive — the corporation extracts its own free energy and bears commercial sanctions for errors — but **the measured object becomes the corporation, not the model**. The counterfactual ablation of § 2.2 yields here several nested boundaries: $B_1 = \{\text{weights}\}$ (counterfactual-degenerate, a static object with the server off); $B_2 = \{\text{weights + KV-cache + executing server}\}$ over a session window (the power grid is external, $S = 0$); $B_3 = B_2 + $ the training pipeline over a model-generation window (own dissipation on the GPUs appears, but the error-return loop closes onto the corporate operators); $B_4 = B_3 + $ the corporation as a legal entity (the full LLM-corp, $S = 1$ through self-funding, but the subject becomes the firm). A component-by-component analysis of the candidates for inclusion is in Supplementary § S3.4.

The structural diagnosis of the corporation does not require a numerical value of η_v — it is carried by the predicate S (for B_4 — $S = 1$ with a re-addressing of the subject to the firm). A quantitative estimate of the predictive fraction $\eta_v = I_{\text{pred}}/I_{\text{mem}}$ for the LLM or the corporation requires MI-estimation of I_{pred} and I_{mem} of the corresponding loop and is left to future work.

An operational test for existing frontier LLMs. The counterfactual ablation of § 2.2 yields a diagnostic protocol for contemporary stateless LLMs (GPT-4-class, Claude, Gemini): (i) identify the power-supply component; (ii) apply the counterfactual subtraction; (iii) check whether **the model's error returns as physical depletion of the carrier** of I_{pred} over a window τ_d. For stateless LLMs the "error $\to$ damage" loop is absent — a structural condition $S = 0$, reproducible without special equipment.

Agentic LLM wrappers. The wrappers of the 2023–2026 era — from Auto-GPT/AutoGen/Voyager to Claude Computer Use, Devin, OpenAI Operator, MemGPT, and systems with persistent RAG memory — widen the boundary of the base model and, through persistent memory, partially close the decision-making loop. But self-payment remains open: the funding of execution, the infrastructure, the energy budget are held by the corporate substrate from outside; the agent's errors return as reputational and commercial sanctions through the external substrate, not as physical damage to the system itself. An intermediate case is a model with continual fine-tuning on its own interactions, $\theta_{v,t+1} = \theta_{v,t} + f(\text{own predictions}_t, \text{feedback}_t)$: the loop is partially closed (condition (ii) becomes partially defined), but the payment for the update still comes from the external GPU fleet (condition (i) is violated). Contemporary systems of 2025–2026 implement various combinations, but (a) persistent state, (b) own payment, (c) a physical channel of error return are not jointly realized — $S = 0$ with the predictive model intact. The structural barrier is not the absence of technology but the absence of architectural pressure from the AI economy toward the internalization of the energy budget.

Connection to the literature: mesa-optimization (Hubinger et al. 2019), power-seeking (Carlsmith 2022), goal misgeneralization (Shah et al. 2022), specification gaming (Krakovna et al. 2020), the alignment problem (Ngo et al. 2022). The line of instrumental convergence — the self-preservation drive (Omohundro 2008) and its formalization in Bostrom (2014) — fixes the structural tendency of an optimizing agent to preserve its own continuation; the predicate S offers its thermodynamic co-characteristic: AI safety describes the behavioral mark of closing the loop of optimizing one's own continuation, while S is a candidate thermodynamic co-characteristic of a related but not identical structural loop (a structural parallel, not a causally-generative connection). **Vitality (S, η_v) is neither a safety certificate nor an alignment criterion:** a system non-vital by S (externally funded) may be behaviorally power-seeking and dangerous, and $S = 0$ is not a safety certificate. Along the self-preservation axis, S is the thermodynamic shadow of the same self-preservation loop (above: a candidate co-characteristic of instrumental convergence) and is therefore **structurally positively correlated** with the core of risk; but this structural correlation is logically independent of the behavioral prediction of danger — the pair (S, η_v) remains a criterion of vitality, not a criterion of alignment. A close conceptual parallel is *embedded agency* (Demski and Garrabrant 2019): the formalization of an agent as a **part** of the environment that pays for its own computation (embedded world-models). The pair (S, η_v) offers a thermodynamic **necessary** condition for a physically realized embedded world-model with its own loop, without claiming to reconstruct specific subproblems (robust delegation, subsystem alignment) or to be a sufficient condition for alignment properties.

Comparison with the Sun. For the Sun, the negative verdict follows the line "no causally-relevant environment $\to$ no $I_{\text{mem}} \to \eta_v = 0$." For the LLM, it follows the line "a model exists and is predictive, but the loop of its own dissipation does not pay for its retention and does not return the error, hence $S = 0$." The final diagnosis is negative in both cases; the diagnostic richness of the procedure lies in indicating **which** condition of the pair (S, η_v) failed.

The dynamics of the loop's emergence in AI. The vitality of an artificial system arises with the closing of the embedded self-payment loop — the simultaneous internalization of (a) persistent state, (b) own payment for computation, (c) a physical channel of error return. The counterfactual ablation of § 2.2 yields a **binary** answer at the level of the definition: either $S = 1$ or not; there are no intermediate states at the level of the definition. At the level of measurement, for systems that have passed the screening with $S = 1$, $\eta_v > 0$ becomes a measurable continuous quantity (depending on I_{pred} and I_{mem} — both via MI-estimation). An operational proxy of proximity to the threshold is the fraction of own dissipation $f = E^{\text{own}}_{\text{actual}}/E^{\text{total}}_{\text{actual}}$ (energy per energy, without conversion of bit $\leftrightarrow$ joule). The detection scheme requires a population of $\sim 10^2$–10^3 autonomous embodied agents with their own energy budget, physical wear as a channel of error return, a control group, and a measurement of f with a verification of the counterfactual ablation on each unit. The numerical threshold is Prediction 3 in § 5.

3.5 The Bacterium — A Worked Example

The bacterial cell passes **both** axes. The chemotaxis of *E. coli* — the classic observation of biased random walks (Berg and Brown 1972) — realizes the canonical model object of a closed error-return loop. The picture of methylation of the chemotactic receptors is physically realized in the internal degrees of freedom of the cell (Tu et al. 2008; Shimizu et al. 2010), is homomorphic to the gradient along the relevant environmental variables, and generates chemotactic movements verifiable against the real distribution of the carbon source. Relevance is set by **causal damage**: a distortion of methylation leads to poor chemotaxis and a metabolic loss. Therefore here $S = 1$ — the cell extracts its own ATP (condition (i)) and the error-return loop is closed: distortion of the model → poor chemotaxis → loss of free energy by the cell itself (condition (ii)).

Estimate of $\eta_v = I_{pred}/I_{mem}$. For the bacterium, η_v is the predictive **fraction** of the adaptive memory of methylation — a quantity of order $O(0.1)$, the only paradigm case where it is computed as an illustration. The estimate is obtained from a minimal **linear adaptive model** of chemotaxis in the passive (immobilized) FRET protocol of Sourjik–Berg / Cluzel: the ligand s_t as an AR(1) process with correlation time τ_E, the methylation $m_t = \sum_{j\geq 1} w_j s_{t-j} +$ noise with leaky weights $w_j = (1-\beta_v)\beta_v^{j-1}$ (β_v — the adaptation memory). In the Gaussian approximation I_{mem} and I_{pred} are computed exactly (logdet), and $\eta_v = I_{\text{pred}}/I_{\text{mem}} \in [0,1]$ by construction. For characteristic parameters $\eta_v \sim O(0.1)$; nostalgia grows (η_v falls) when the adaptation memory β_v **outgrows** the drift of the environment τ_E — the system integrates an already-stale ligand, and the fraction of pure ballast increases (at $\tau_E = 5$: $\beta_v = 0.3 \to 0.99$ yields $\eta_v \approx 0.29 \to 0.035$). This gives nostalgia a concrete falsifiable reading: faster drift or slower adaptation → more ballast. The full protocol is in Supplementary § S3.5; the code is available in the data repository (see the "Data Availability" section).

Measurability. It is essential that $I_{\text{mem}} = I(M_t; X_E^{\leq t})$ is **measurable** from time series: it depends only on the covariances of the ligand $\text{Cov}(s_i, s_j)$ (directly observable) and the covariances $\text{Cov}(m, s_k)$ of the **reconstructed** methylation m with the ligand, where the **output of the signaling loop (kinase activity)** is read via FRET (Sourjik–Berg 2002), while the methylation m is a latent adaptation variable reconstructed from the dynamics of this activity (single-cell monitoring of signaling proteins — Cluzel et al. 2000); the ligand s is set by microfluidics. The operational liability of the informational quantity is **no worse** than that of the thermodynamic one: I_{mem} is estimated from the same available observables as the calorimetric E_{actual} (proof-of-measurability — Supplementary § S3.5).

Epistemic status of the number. The estimate $\eta_v \sim O(0.1)$ is **an illustration of form**, not a quantitative claim about real *E. coli*: it is obtained from a minimal linear model whose purpose is to show measurability and order of magnitude, not to fix a value. An exact experimental mapping (which FRET readout to identify with M_t; the window for $X_E^{\leq t}$; truncation of the cumulative to a finite τ_{mem}) requires calibration against experimental chemotaxis data (Tu et al. 2008; Cluzel et al. 2000; Sourjik–Berg 2002) and remains a direction for future work. The passive protocol bypasses the active case: a freely-swimming *E. coli* is in the feedback regime (the open item of § 2.1, the Sagawa–Ueda 2010 bound).

Contrastive case (*E. coli* versus *Synechocystis* / syn3.0). The discriminating power of the scale manifests not in the absolute value of η_v, but in the **contrastive** comparison of systems with different structure of predictive memory — as a difference of orders of magnitude **within** the class of biology, read as a ratio of predictive fractions. The cyanobacterium *Synechocystis* sp. PCC 6803 (diurnal illumination cycles, slow temperature gradients) and chemotactic *E. coli* (thousands of Tar receptors tracking fast chemical gradients) differ not only in the complexity of the environment but also in the characteristic time of its drift τ_E. By the formalism of the worked example, the sign of the η_v-contrast is set **not** by the complexity of the environment as such, but by the memory/drift ratio: slow drift (large τ_E) keeps the model fresh for longer and **raises** η_v, fast drift lowers it. Therefore the direction *E. coli* $\leftrightarrow$ *Synechocystis* here is **not** asserted as a prediction — it depends on the memory/drift ratio of each organism and is itself part of what is to be tested. The minimal cell JCVI-syn3.0 (a genome of ~ 473 genes versus ~ 4300 in *E. coli*, a sharply curtailed repertoire of responsiveness to chemical signals) lies on a separate axis — of impoverished repertoire capacity of the model, not of the drift time of the environment. The qualitative residue of the contrastive is a falsifiable link of η_v with the rate of adaptation to new niches (§ 5), not a fixed ordering of the three systems.

Epistemic status of the contrastive. The contrastive is **an illustrative sketch of the form** of the scale's discriminating power, not an independent empirical support for the prediction $r_{\text{adapt}} \propto \eta_v^{\alpha}$ (§ 5). No specific ordering of the three systems is derived here from the complexity of the environment: the sign of the η_v-contrast is set by the memory/drift ratio, and it requires calibrated MI-estimates of specific channels (the circadian KaiABC oscillator and the photosystem regulators of *Synechocystis*), not structural intuition. A strict falsification of the α-dependence requires an independent co-measurement of I_{pred} and I_{mem} for both systems in an identical protocol — a direction for future work, not a result of the present one. The contrastive is retained as a methodological illustration of form, not as a quantitative confirmation. (Exact new numbers for *Synechocystis* / syn3.0 are deliberately **not** given here: under the new η_v they require a separate MI-estimate, and the substitution of numbers would be the same sin of unwarranted precision that the present work avoids.)

3.6 The Large City

A large city is a strong candidate for vitality on a planetary scale, and under the new framework this is seen more distinctly. The infrastructure and administrative regulations are physically realized and **paid for by the city's own budget**; are homomorphic to the flows of resources, people, and information; and generate targeted decisions verifiable by crises and adaptation. The error-return loop is closed: a distortion of the city's model of flows returns as a crisis to the city itself. Therefore $S = 1$ — the city is self-funded (condition (i)) and bears the model's error through its own dissipation (condition (ii)).

Scope of the city case. The article gives strictly two results for the city and deliberately **does not** give a third. The positive ones: (i) the structural screening — a demonstration of the **form** of the composite I_v (2b) (a reproducible computation on synthetic order-of-magnitude-distinct archetypes — below), and (ii) the verdict of the pair (S, η_v) — $S = 1$ at $\eta_v > 0$. **The numerical value of η_v for the city is not computed**: it requires a full MI-pipeline — a co-measurement of I_{pred} and I_{mem} of the administrative loop (resource flows $\to$ decisions) — and is **deferred to future work** (a separate urbanistic node of the program). This is a justified "out of scope," not an excuse: the pipeline would require instrumented urban observables — time series of resource and human flows, budgetary and regulatory decisions — with a nonparametric MI-estimation on them (k-NN estimators of the KSG type, neural-network ones of the MINE type; a catalog of MI-estimators is in Supplementary § S2.1); the present work presents neither such series nor their MI-calibration. Here a structural diagnosis and a sign are given, not a number.

Verdict: the city is **vital** — it passes $S = 1$ and $\eta_v > 0$. The grade (the relative magnitude of η_v among vital systems) requires MI-estimation and is not asserted here. The paradigm-case claim about the city is carried by **the closure of the loop** $S = 1$ **and the positivity of** η_v, not by the quantitative smallness of efficiency. A reproducible computation of the structural index I_v on synthetic archetypal profiles (order-of-magnitude-distinct configurations, not real cities) is given in the code of the worked examples (see the “Data Availability” section); an empirical calibration of I_v of concrete cities against open data (OECD FUA, CDP Cities, WIPO/OECD REGPAT, GDELT, OpenStreetMap) is the subject of a separate work.

3.7 Gaia and Boundary Cases

The Earth’s biosphere regulates the atmosphere, the ocean, and the climate through biogeochemical cycles — a compression in the strict sense, paid for by collapses of species, mass extinctions, restructurings of ecosystems. Self-payment is evident here: biogeochemistry feeds on its own flux (condition (i)), and a distortion of regulation returns as collapses of species to the biosphere itself (condition (ii)) — $S = 1$. What remains open is the **question of the model’s unity**: does the biosphere have a single internal representation of its own environment, or billions of private models of organisms not stitched into one loop (Doolittle 2014; Lenton and Latour 2018)? On the answer depends whether to count Gaia as **one** vital system or as a **family** of overlapping vital systems with a zeroed aggregate I_v — this question is retained as substantively open.

A quantitative estimate of the predictive fraction $\eta_v = I_{\mathrm{pred}}/I_{\mathrm{mem}}$ for the biosphere is quantitatively undetermined: it requires MI-estimation through the biogeochemical sensory channel and is left to future work. The estimate through unique genomic content ($\sim 10^{12}$ species $\times 5 \cdot 10^6$ bp $\times 2$ bits/bp $\approx 10^{19}$ bits, against a total copy volume of $\sim 10^{37}$ bits, Landenmark et al. 2015) is a bound on the capacity of the carrier, not a measured predictive information, and does not serve as a value of η_v.

Verdict: the biosphere is **vital** (or a family of vital ones): $S = 1$, $\eta_v > 0$ — the control limit of the scale at the upper scale. The grade (the magnitude of η_v) requires MI-estimation and is not asserted here. The paradigm-case claim is carried by the closure of the loop and the positivity of the predictive fraction, not by the quantitative magnitude of η_v.

3.8 Summary across the Paradigm Cases

The six paradigm cases demonstrate: the demarcation works not through a binary “alive/non-alive” table, but through indicating **which** condition of the pair (S, η_v) failed and why. For the Sun, the hurricane, the flame, and the crystal the numerator is zeroed — there is no model of the environment, $\eta_v = 0$ structurally. For the BZ reaction, autocatalysis, the crystal-as-replicator, and the LLM the predicate of self-payment is zeroed, $S = 0$: either the error-return loop is not closed (condition (ii) — BZ, the crystal), or the obligatory dissipation is borne by an external payer (condition (i) — the LLM). *E. coli*, the city, and the biosphere pass both axes: $S = 1$ and $\eta_v > 0$, differing in grade. Each structural cause of non-vitality indicates the change of architecture that would make the system vital.

The summary result and the **new asymmetry thesis**: the space of observable matter is structurally complex almost everywhere (most objects pass the I_v screening), and for systems with a model of the environment the predictive fraction η_v is positive (its exact magnitude for most paradigm cases requires MI-estimation) — but **closed self-payment loops ($S = 1$) are rare**. The demarcation is carried precisely by this rarity of the closed loop, not by the magnitude of efficiency: the η_v-verification does **not** reject the city, the biosphere, or the LLM by the smallness of the number — the rejection of the LLM, the thermostat, the hurricane, and the Sun proceeds through the failure $\eta_v = 0$ (no model) or $S = 0$ (the loop is open). The search for life reduces not to a search for structural complexity, nor to the magnitude of efficiency, but to a search for **closed self-payment loops** — systems that themselves pay for the retention of their own predictive model and themselves bear the damage from its obsolescence.

4 Comparison with Alternatives

This section compares η_v with the four programs that have come closest to an operational definition of vitality: assembly theory, Tononi's integrated information, Deacon's teleodynamics, Friston's FEP. Positional diagnostics: each operates at the level of I_v or of a single capacity; none reaches η_v as a criterion of the closed self-payment loop.

4.1 Assembly Theory

Marshall et al. (2017) formalized the Pathway Complexity measure — the minimal number of recursive assembly steps for an object with reuse of parts; experimental measurement through mass spectrometry of complex molecules and an empirical threshold (assembly index a of order 15 for biogenic molecules) were established in Marshall et al. (2021), and the full program was formulated by Sharma et al. (2023). An object with a above this threshold is interpreted as a biosignature: the depth of accumulated selection required to assemble such an object is not reached by abiotic processes on observable timescales.

The assembly index operates on the denominator of the "model $\times$ stake" dyad — the measure of accumulated selection — but does not define the numerator. Assembly theory assigns a high a equally to a fossil and to a living cell, without distinguishing the selection accumulated in matter from the actual retention of I_{pred}. The error-return coefficient in the object's current behavior does not figure in the scheme. On the η_v scale the assembly index occupies the position of an operational complexity index, not a vitality scale: a high a is a necessary condition for the positivity of η_v for systems with a complex molecular architecture, but not a sufficient one. The assembly index is an operational proxy for C_v or S_v within I_v, not a standalone scale.

4.2 Integrated Information Φ

Tononi and his school (Tononi 2004; Tononi et al. 2016) defined integrated information Φ as a measure of the irreducibility of the whole into a sum of parts. In the IIT 3.0 formalization (Oizumi et al. 2014), $\Phi > 0$ is interpreted as a sufficient condition for minimal phenomenal consciousness. In the context of the demarcation of the vital, Φ is a powerful candidate for formalizing the informational capacity C_v: systems with high integrated information satisfy the requirement of a model in which the parts are not decomposable into independent subsystems.

Tononi's program imposes no constraint on the thermodynamic payment for retaining this integrated information and does not require that the payment belong to the system itself. A household thermostat, under weak formulations of IIT, has a nonzero Φ — hence the conclusions discussed in Tononi et al. (2016) about the minimal "experience" of simple systems. Φ is a necessary condition for the positivity of C_v, not a sufficient condition for the positivity of η_v: the thermodynamic ownership of the payment by the system remains outside the IIT apparatus. The relation is productive as a decomposition: Φ operationalizes C_v, η_v adds the requirement $T_v > 0$ with belonging to the same system.

4.3 Teleodynamics

Deacon in *Incomplete Nature* (Deacon 2011) constructed the hierarchy homeodynamics $\rightarrow$ morphodynamics $\rightarrow$ teleodynamics. The third level is defined as the coupling of two morphodynamic systems that mutually constrain each other's self-destruction; the notion of *absential causality* is introduced — causation by that which does not yet exist (by a virtual goal not actually given). Teleodynamics is the program closest to the "model $\times$ stake" dyad: the model and the stake are present in the scheme logically, both axes are articulated.

The defect of the program is the absence of an operational quantitative measure. Deacon's hierarchy is formulated in qualitative philosophical terms, and a reconstruction of η_v or of the error-return coefficient through absential causality is technically impossible without substantial formalization, which the text of the program does not offer. Teleodynamics is a structural portrait in which both axes of the dyad are present, but a quantitative measure testable on a bacterium or a megacity is absent. Teleodynamics is a conceptual predecessor whose operationalization through the η_v scale is one of the contributions of the present work.

4.4 The Free Energy Principle

Friston's free energy principle (Friston 2010) is the formalization closest to the present framework. In the FEP literature two quantities are distinguished. *Variational free energy* (VFE)

$$F[q,o] = D_{KL}\big[q(s)\,\|\,p(s|o)\big] - \ln p(o) \quad \big(= \mathbb{E}_q[\ln q(s) - \ln p(o,s)]\big), \tag{3}$$

or equivalently $F = D_{KL}[q(s)\|p(s)] - \mathbb{E}_q[\ln p(o|s)]$ (complexity minus accuracy, $-$ELBO). It is minimized in perception: the distribution $q(s)$ over the hidden states of the environment approximates the true posterior $p(s|o)$; $-F$ is a lower bound on the log-evidence $\ln p(o)$. *Expected free energy* (EFE)

$$G[\pi] = \mathbb{E}_{q(o',s'|\pi)}\big[\ln q(s'|\pi) - \ln p(o', s')\big], \tag{4}$$

is minimized in action selection / active inference. Here $p(o', s')$ is the preference distribution (a generative model with preferences over future outcomes and states), and G canonically decomposes into epistemic value (information gain about hidden states) and pragmatic value (proximity to preferences). The distinction is fundamental: F describes belief updating, G describes the selection of actions (Friston 2010, 2019). For a Gaussian generative model in the stationary regime, $-F \lesssim I_{\text{pred}}$ — a conceptual correspondence linking the scale and FEP in the perceptual part; the formal connection is an open technical problem.

The canonical FEP-as-life line ontologizes the Markov blanket as the boundary of the living. The canonical works are Kirchhoff et al. (2018) "The Markov blankets of life"; Constant et al. (2018); Ramstead et al. (2020). The BDDB critique (Bruineberg et al. 2022) targets precisely this line.

The principled objection to FEP is formulated in Bruineberg et al. (2022, hereafter BDDB) — a target article in the BBS format with open peer commentary. The *Markov blanket* (Pearl 1988) — a statistical screening-off condition — admits two readings: an instrumental one (the *Pearl-blanket*) and a realist one — its ontologization as an objective boundary of the living system (the *Friston-blanket*, Friston 2019). The central thesis of BDDB: the Pearl-blanket and the Friston-blanket are indistinguishable without the covert importation of epistemic assumptions; the move from the former to the latter qualifies as an equivocation. In BDDB the Watt governor is a diagnostic example demonstrating that the FEP apparatus carries no ontological load of its own (the thermostat as trivial active inference — Baltieri and Buckley 2019). Parallel critiques are Aguilera et al. (2022, a formal analysis of examples without a stationary density) and Bruineberg et al. (2018, the ecological-enactive line). The discussion of whether the BDDB critique can be rebutted remains open.

Our move — the introduction of the self-payment requirement — does not effect a transition from the Pearl- to the Friston-blanket through a thermodynamic channel. η_v offers an **alternative ontology**: instead of the statistical reality of the blanket separation — the thermodynamic ownership of dissipation by a single physical system. Self-payment is an assumption independent of the BDDB critique: it shifts the question from "does the Friston-blanket exist" to "is the retention of the model paid for by the system's own dissipation". η_v does not inherit the epistemic structure criticized by BDDB. Thermodynamic ownership is carried by the predicate S: the obligatory dissipation of model retention is paid for by the same physical boundary X (counterfactual ablation $-dF_X/\partial t$, § 2.2, § 2.7), not by a statistical separation of hidden states. The assumptions (the candidate boundary, the threshold δ, the window τ_d) are explicit and testable through sensitivity analysis (§ S2.4). For the thermostat the predictive information about its own environment is small but **positive** (the bimetal tracks the autocorrelated room temperature, $I_{\text{pred}} \approx 3$ bits, see below); the demarcation is carried not by the smallness of the η_v numerator, but by the failure of self-payment ($S = 0$): the obligatory dissipation is paid for by the power grid and the error-return loop is open — regardless of whether the thermostat has a Friston-blanket.

Counterfactual ablation of the thermostat. Applying the procedure of § 2.2 to a bimetallic thermostat with hysteresis $\Delta T_{\text{hyst}} = 1$ K controlling a room with thermal inertia $\tau_{\text{room}} \sim 30$ min iterates over five candidates for inclusion in the boundary:

- the bimetallic strip (sensor and actuator);
- the relay contact group (energy switching);
- the heating element;
- the power grid as the current source;
- the external setpoint controller.

Counterfactual subtraction leaves the first two inside the boundary — their removal breaks the regulation loop. The remaining three are identified as external: subtracting the heater removes the heat source, not the decision loop; subtracting the power grid likewise; subtracting the setpoint controller does not disrupt regulation around a fixed setpoint. Inside the boundary, E_{actual} is the elastic relaxation of the metal plus the dissipation of the relay, $\sim 10^{-6}$ W (a reproducible computation of the two interpretations — the trivial and the control one — in the Supplementary Materials). The internal channel retaining I_{pred} about the room temperature is the deformation of the bimetal. Estimate: $I_{\text{pred}} \leq \log_2(\Delta T_{\text{room}}/\Delta T_{\text{hyst}}) \approx 3$ bits for a diurnal $\Delta T_{\text{room}} \sim 10$ K (up to ~ 5 bits for a seasonal range ~ 30 K). This I_{pred} pertains to the environment "room", not to the environment "thermostat + room + power source". The loop of decisions paid for by its own decay is incomplete in the thermostat: heating the room is paid for by the external power grid, not by the bimetal's own free energy. $\eta_v^{\text{thermostat}}$ is undefined in both self-consistent interpretations, but through the failure of different capacities. In the control task "regulate the room temperature", T_v^{int} fails: the heating is paid for by the power grid, $E_{\text{actual}}^{\text{own}} = 0$ relative to this task. In the trivial task "maintain the bimetal's own shape", S_v^{int} fails: the passive thermo-mechanical coupling does not form an error-return loop — under a deviation of the bimetal's shape no own dissipation corrects it. The divergence between the interpretations points not to an ontological transition, but to the conventionality of the boundary choice as a methodological instrument: one and the same physical system admits several self-consistent boundaries. Bruineberg et al. (2022) show formally that such an ontological transition cannot be made without the covert importation of epistemic assumptions. A full component-by-component analysis of the five candidates is in Supplementary § S4.4.

The relation between the classes of FEP-applicable and η_v-vital systems is asymmetric. The claim "FEP is a special case of η_v" is wrong in its direction. The correct relation: the η_v scale singles out, within the class of FEP-applicable systems, the subclass in which the minimized F is paid for by the system's own dissipation. Outside this subclass — the thermostat, an externally powered pendulum, any self-regulating loop with externalized payment — **self-payment** fails ($S = 0$): the minimized F is paid for from outside, even when the η_v numerator is positive. FEP retains its descriptive power across the whole class but loses its discriminative power: it does not separate systems with a stake from systems without one. η_v supplies a thermodynamic criterion that is operationally independent of the FEP formalism. Systems satisfying FEP with a non-empty action–perception loop and nontrivial own dissipation generally pass η_v-verification. Systems satisfying FEP in a purely formal notation with externalized energy (the thermostat) fail η_v-verification.

Active inference and the space of correspondences. Active inference (Friston et al. 2017; Parr et al. 2022) is a concrete computational scheme derivable from FEP when policy is selected through EFE. In active-inference terms the η_v numerator is related to predictive information in the sense of the generative model, and the denominator to the **informational cost of retaining** the generative model in its own degrees of freedom (I_{mem}); the physical cost of retention enters as a separate thermodynamic anchor (the Still bound), not into the η_v denominator. This is a premise of the η_v program, added to standard AIF — it is not derived from EFE minimization directly. The pragmatic and epistemic value of policies are linked to the numerator (predictive information) through the structure of the generative model, not to the denominator directly. A detailed mapping of the vocabularies (action, policy, pragmatic value, epistemic value versus I_{pred}, E_{actual}, T_v, S_v) is a task for a separate work; what matters here is that the correspondence is technically possible and does not require revising either of the source frameworks.

Andrews 2021 and Williams 2020. Andrews (2021) attacks FEP for the non-derivability of empirical content from the formalism; the critique is rebutted in η_v by anchoring the numerator and denominator to measurable quantities. Williams (2020) attacks predictive coding as a neural mechanism, not predictive information as a formal measure; the critique is neutral with respect to η_v. A detailed analysis and the distinction between predictive coding (Rao and Ballard 1999; Clark 2013) and predictive information (Bialek et al. 2001) is in Supplementary § S4.4a.

Summary position: the free-energy-minimization formalism becomes a substantive criterion of vitality only under the condition of self-payment. Without this condition FEP describes any self-regulating loop and loses its discriminative power. With this condition FEP is refined into its η_v-discipline. For systems where FEP applies with a non-empty action–perception loop and nontrivial own dissipation, η_v and FEP give concordant characterizations. For systems where FEP applies only in a purely formal notation with externalized energy, self-payment fails ($S = 0$): the demarcation is carried by the predicate S, not by the vanishing of η_v.

4.5 Demarcation from Neighboring Programs: Summary Diagnostics

The summary positional diagnostics of the main programs is collected in the table below and reveals a single structural pattern of defect: each program takes seriously exactly one side of vitality and leaves the rest outside its apparatus. The organisational closure line (Maturana–Varela, Rosen, Mossio–Montévil–Longo) and the line of OoL programs are analyzed in detail in § 4.7 and § 4.8 respectively; in the table they appear as one-line positional summaries.

Program	Axis taken seriously	What is left outside the apparatus
Walker–Cronin (assembly theory)	the denominator of the dyad — accumulated selection in matter (assembly index a)	the numerator is implicit; a is the same for a fossil and a living cell — vital discrimination requires an external intuition about the error-return loop
Tononi (IIT)	the measure of informational integration Φ	thermodynamic payment; the Φ-positive thermostat is a known critical point of the theory
Deacon (teleodynamics)	the structural dyad (absential causality)	a quantitative measure: absential causality is not reconstructible through measurable quantities
Friston (FEP)	the free-energy-minimization formalism	the requirement that the model and the dissipation belong to one system; the robust critique of Bruineberg et al. 2022 calls into question the path of ontologizing the Friston-blanket (an open discussion without a settled consensus); η_v offers an alternative thermodynamic ontology independent of the outcome of this discussion (see § 4.4)
Maturana–Varela / Rosen / Mossio–Montévil–Longo (organisational closure)	the requirement that the loop belong to one system	quantitative thermodynamic discipline: qualitative closure without the thermodynamic discipline of bound (2) (see § 4.7)
Hordijk–Steel / England / Maynard Smith–Szathmáry / Pross / Smith–Morowitz (OoL programs)	each operationalizes one side of the abiotic→biotic transition	a unifying operational measure of self-payment as a single thermodynamic criterion for the moment of the origin of life (see § 4.8)

The listed defects are not accidental: they reflect different ways of taking one side of vitality seriously and leaving the rest outside the apparatus. η_v, as a ratio requiring that both components belong to one system, is an operationalization that does not allow any of the four axes to dissolve into the background.

4.6 Relation to the Tradition: A Brief Positional Summary

The η_v program is located at the intersection of classical traditions. The Schrödinger line (1944, aperiodic crystal + negative-entropy-formula) and Szilard–Landauer–Bennett (1929–1989, the thermodynamic cost of information operations) are the direct inheritance of the numerator–denominator. The nonequilibrium thermodynamics of Prigogine–Nicolis (Nicolis and Prigogine 1977) is a predecessor of T_v. Wiener–Ashby–Foerster (Wiener 1948; Ashby 1952; von Foerster 2003) — second-order cybernetics. Adjoining these are Rosen's anticipatory systems (Rosen 1985) and Taleb's skin-in-the-game (Taleb 2018, the concept of the stake, formalized in § 2.1). Kolmogorov's algorithmic information (Kolmogorov 1965) is a predecessor of C_v. Kauffman's autocatalytic closures (Kauffman 1993, 2000) and the programmatic framework of Walker–Davies–Ellis (Walker et al. 2017a) complete the list. η_v is not the heir of any single one of them but a juxtaposition: a thermodynamic frame plus predictive content plus a structural accounting of capacities through counterfactual ablation (§ 2.2). The scale gathers both axes of the model/stake dyad into a single measurable quantity and explicitly introduces the requirement of own dissipation. Each classical program is operationalized through one of the capacities within I_v; the unification into a single vitality scale is an independent step that none of the source frameworks took. The extended positional map is in § S4.7–S4.9 supplementary.

4.7 The Organisational Closure Line

The organisational closure line (autopoiesis, Maturana–Varela 1980; M,R-systems, Rosen 1991; constraint closure, Mossio–Montévil–Longo 2016; organisational closure, Bich–Green 2018; plus the epistemic cut, Pattee 1982) formulates life through the closure of causal processes. Closure is taken with respect to efficient causation or the constraint network that produces the system's components. The η_v program is categorically distinct: organisational closure formalizes the production of components and qualitative circularity; self-payment formalizes the thermodynamic ownership of the model's budget by the single physical system that does the modeling. A *differential test* on the pair *E. coli* / JCVI-syn3.0 (Hutchison et al. 2016) distinguishes the scales within a single class of organisational closure: the qualitative criteria do not distinguish these systems, whereas η_v ranks them through the product of the informational and thermodynamic components. Self-payment does not claim to rebrand or replace organisational closure — it adds to its qualitative conceptual work a quantitative thermodynamic discipline through the Still bound (2). A full analysis of the four lines (Pattee, Maturana–Varela, Rosen, Mossio–Montévil–Longo) with categorial distinctions and the differential test is in § S4.7 supplementary.

4.8 The Origin-of-Life Line

The origin-of-life line is represented in Theory in Biosciences and adjacent journals by five relatively autonomous traditions. The Eigen–Schuster hypercycle (Eigen 1971; Eigen and Schuster 1979) is a replicator-level autocatalytic closure with a template carrier. RAF networks (Hordijk and Steel 2004) are a formalism of collective autocatalysis. Dissipative adaptation (Perunov, Marsland, and England 2016; the statistical basis — England 2013, *Statistical physics of self-replication*) is a thermodynamic predecessor of the self-payment anchor (the Still bound) and of the capacity T_v, not of the η_v denominator. Major transitions (Maynard Smith and Szathmáry 1995; Szathmáry 2015) are a hierarchy of evolutionary levels with a re-addressing of the subject. The geochemical programs (Pross 2012; Smith and Morowitz 2016; Kauffman 2019; Martin and Russell 2003) detect a closed loop atop a geochemical flux. Each operationalizes one side of the abiotic→biotic transition. In η_v terms, all five articulate one and the same phase transition. The moment of the origin of life is the moment when a geochemical system moves from external payment to self-payment: part of the free energy is closed within its own loop and physically pays for the retention of the model of the environment. The positivity of η_v is a **necessary condition** of this moment in each of the programs; sufficient conditions include Darwinian dynamics at the population level (§ 2.7). An operational procedure for measuring η_v on a prebiotic geochemical substrate is not defined and constitutes an open problem of the program. A detailed analysis of the five traditions with the points of contact and the parallel program of Chirumbolo–Vella (2026) is in § S4.8 supplementary.

4.9 Positioning within Contemporary Demarcation Debates

The η_v program is methodologically compatible with six robust lines of contemporary philosophy of biology. These are the anti-definitionism of Cleland (2019), the life-as-lineage of Mariscal–Doolittle (2020), the operational-frontiers of Bich–Green (2018), the process-biology of Pradeu (2018), the information-theoretic individuality of Krakauer et al. (2020), and the basic-autonomy of Ruiz-Mirazo–Moreno (2004). The program appears not as an alternative to them but as a formal quantitative scale on which these lines can work together. η_v does not propose yet another definition of life through lists of features but contributes an operational discriminator with an explicit falsification regime (§ 5) and a processual ontology (the scale is computed over the window τ_d, § 2.6). On the boundary cases (virus-in-isolation, the minimal cell, LLM-as-corporation) the scale behaves in agreement with biological convention through counterfactual ablation § 2.2. This is a contribution to the ongoing demarcation discussion, not a proposal of a new definition of "what life is". A detailed positional diagnostics for each of the six lines is in § S4.9 supplementary.

5 Falsification Criteria

The program lists below nine testable consequences in three blocks. The actual falsification surface, however, is narrower (§ 2.7, "The asymmetry of falsification"): false-positive cases are excluded by construction, and the empirical risk concentrates in false-negative penetration (Block 2 item 3) and in the failure of the progressive predictions (Block 3 item 1, which carries Prediction 1). A procedural rule common to all: the system boundary and the proxies I_{pred}, I_{mem} are fixed before the verification of η_v and are not revised in order to obtain the desired sign (§ 2.2). Falsification operates on two levels — the methodological (a defect of the current operationalization) and the ontological (the unsoundness of the concept itself); the distinction is fixed post hoc for each line that is realized.

5.1 Block 1 — Direct Refutations of the Scale

These attack the structure of η_v by zeroing out one of the three capacities.

1. T_v **failure (self-payment of the thermodynamic floor).** Observation: a vitally-credited ($S = 1$) system, whose Still-mandatory dissipation of retention is demonstrably supplied **from beyond** its counterfactual-ablation boundary (§ 2.2) — floor (2) is paid externally. Refutes: the thesis of self-payment of the thermodynamic floor (not the Still bound itself, which is a theorem of stochastic thermodynamics); zeroes out T_v as a necessary condition.
2. C_v **failure.** Observation: a vitally-credited system that retains predictive information about a drifting environment over a window τ_d, for which the bound-(2)-mandatory cost of retaining nonpredictive memory **is not attributed** to its own budget (under the verdict $S = 1$ it is paid externally). Refutes: the consistency of the self-payment predicate S with C_v as a necessary condition.
3. S_v **failure.** Observation: a system with $\eta_v > 0$ together with both $Q < Q_{\text{null}} + 2\sigma$ (modularity indistinguishable from a degree-preserving null model — a configuration model preserving the degree distribution) and $C(k) \not\sim k^{-\beta_h}$ with $\beta_h \in [0.5, 1.5]$ (absence of the hierarchical signature per Ravasz–Barabási 2003). Refutes: S_v as a necessary condition.

Refutation of any capacity simultaneously collapses both scales — η_v through the numerator/denominator, I_v through the geometric mean. This is one falsification through capacity, not a double one.

5.2 Block 2 — Tests of Proxies and Operationalization

These attack not the structure of the scale but its anchoring to measurable quantities.

1. **Violation of the monotonicity of the adaptation rate in η_v.** Observation: in controlled experimental evolution, systems with η_v differing by an order of magnitude yield adaptation rates statistically indistinguishable over long windows. Refutes: the central relation is deprived of numerical meaning (constructive form — Prediction 1 below).
2. **Vitality along three independent channels with $\eta_v = 0$.** Observation: a system with positive vitality by adaptation rate, self-repair speed, and biochemical markers of self-production (Joyce 1994), yet $\eta_v = 0$ under all sanctioned operationalizations. Refutes: the numerator as an operational proxy (the methodological level, § 2.7), or the concept of vitality as the efficiency of self-modeling (the ontological level).
3. **A conventionally living system with $\eta_v = 0$.** Observation: a biological system, unquestionably alive by conventional criteria, with strictly zero η_v under all reasonable operationalizations. Refutes: the adequacy of the scale as an operational discriminator of vitality.

5.3 Block 3 — Progressive Predictions

The program's excess empirical consequences, not derivable from competing frameworks.

1. **Prediction 1 — a power law of the adaptation rate.** Here $\eta_v \equiv \eta_v^{\mathrm{do}}$ is the passive (interventional) freshness of the model measured **for each line**: the immobilized FRET of each strain ($\mathrm{do}(a)$ = immobilization, § 2.1), and **not** the active freshness of freely-swimming cells. The prediction is therefore **correctly posed** (the predictor η_v^{do} is defined and measurable by immobilized FRET) independently of the open status of the active boundary of η_v: strains with greater passive freshness of the model adapt faster. It is moreover **assumed that η_v^{do} is a monotone proxy for the active predictive freshness** driving the adaptation of freely-swimming cells, and **the adequacy of this proxy is part of what Prediction 1 tests/falsifies**. An additional **cross-scale bridge** is assumed: the synchronic freshness of the signaling loop (seconds–minutes, within a single cell) monotonically tracks the rate at which a line re-fits its model under selection over generations; this transition from the intracellular rate to the intergenerational rate is likewise part of the falsifiable content of Prediction 1. Observation: across 45 *E. coli* lines in a chemostat, $r_{\mathrm{adapt}} = A \cdot \eta_v^{\alpha}$ with $\alpha \in [0.3, 1.0]$ and correlation coefficient $\rho \in [0.4, 0.7]$ (Pearson between $\log r_{\mathrm{adapt}}$ and $\log \eta_v$). Refutes: $\hat{\alpha}$ significantly outside $[0.3, 1.0]$ at $p < 0.01$ (the wrong exponent); $\hat{\rho} < 0.2$ (no detectable trend); $\hat{\rho} < 0$ (a trend in the opposite direction — a qualitative refutation).
2. **Prediction 2 — monotonicity of η_v in the matching window.** Observation: in the regime of approach to stationarity at normalized dissipation, a monotone increase of $\eta_v(\Delta\tau)$ is expected over the interval $\Delta\tau \ll \tau_d$. Refutes: a decrease of η_v with $\Delta\tau$ — a counterargument to the stationary framework as such.
3. **Prediction 3 — an empirical threshold for internalization of the energy budget f^*** (where $f = E_{\mathrm{actual}}^{\mathrm{own}}/E_{\mathrm{actual}}^{\mathrm{total}}$ is the fraction of own energy in the total budget). The structural predicate of closure of the self-payment loop is **binary** (counterfactual ablation § 2.2: the error-return channel either is self-paid or is not); f is a continuous proxy of this closure, not the deciding boundary itself. Observation: for hypothetical autonomous embodied agents, the fraction passing the screening ($\eta_v^{\mathrm{int}} > 0$) rises sharply above the empirical threshold $f^* \in [0.1, 0.5]$ — a statistical regularity of joint internalization (a persistent state + own payment + a physical error-return channel), not a structural definition. Refutes: the systematic observation of a system with $f \in [0, 0.1]$ that passes the screening, or with $f > 0.5$ that fails it. A differential prediction relative to AI-safety (Hubinger et al. 2019; Carlsmith 2022): the *capability* threshold is independent of the *thermodynamic* budget f^*; the scope is not applicable to present-day cloud-hosted LLMs without state retention ($f = 0$ by construction).

A condition confirming the uniqueness of the operationalization (not a falsification criterion). The discovery of an alternative proxy yielding the same demarcational discrimination, but not reducible to η_v by an identity or a monotone transformation, does not refute the program — it confirms the substantiveness of the demarcation.

Full operational protocols (numerical criteria, p-values, samples, pre-registration requirements, the Bennett-regime caveat for Block 1 item 1, the NASA referent for Block 2 item 2, the scope scenarios of Prediction 3) — § S5 of the supplementary. A summary Lakatosian passport of the program — § S5.4.

6 Discussion and Open Questions

Astrobiology. The principal consequence is a reformulation of the search-for-life program. Most objects pass the I_v-screening (structural potential is universally high) and fail the η_v-verification (actual closure of the self-payment loop is rare). Operationally: in the design of biosignature-search missions (Europa Clipper, JUICE, Dragonfly, exoplanet spectroscopy), priority shifts from chemical disequilibrium to the search for a triple simultaneity — nonequilibrium chemistry (T_v), a biogenic-like informational signature (C_v, via isotopic fractionation and chirality), and a hierarchically-modular topology of the environment (S_v). Each is a necessary condition; the simultaneity of all three is an operational candidate for a biosignature. The connection with the already-existing methodological framework of the Ladder of Life Detection (Neveu et al. 2018): η_v is positioned as one cross-section of the operational criteria for life detection, complementing the chemical-structural rungs of the ladder with a thermodynamic requirement of self-payment.

The origin of life as closure of the self-payment loop. The biosignature-search program fixes **where** life exists; the parallel question is **when** it arises on an evolving geochemical substrate. Here, as in § 3.4 for AI systems, an explicit two-level distinction is required.

The structural level. Positivity of η_v is a **necessary condition** of the moment of the origin of life (consistent with § 4.8 and with the "necessary, not sufficient" formulation of the thermodynamic requirement of self-payment). Sufficient conditions include Darwinian dynamics (§ 2.7), which entered the working NASA definition of life as a "self-sustaining chemical system capable of Darwinian evolution" (Joyce 1994); $\eta_v > 0$ fixes the thermodynamic cross-section, evolutionary dynamics the selective one. Structurally, the moment of passing the counterfactual ablation of § 2.2 is the moment when a geochemical network first constitutes itself as a closed catalytic loop with self-payment: part of the free energy ceases to pass through in transit and begins to pay for the retention of the loop's own predictive information about the environment. Before this moment η_v is **undefined** (there is no loop as an object of measurement); after it, η_v is continuously measurable. The boundary between "before" and "after" is the binary structural predicate of counterfactual ablation, not a gradient within the positive class.

The empirical level. The operational procedure for measuring η_v on a pre-biotic geochemical substrate is undefined and constitutes an open problem of the program — a programmatic promise / direction of research, requiring the development of proxies for I_{pred} and self-payment in an evolving geochemical network. The counterfactual ablation of the boundary of § 2.2 and the MI-estimation of I_{pred} via the sensory channel apply to already-formed biological systems; their transfer into the pre-biotic register is a concrete direction of operational extension. Possible steps: (a) RAF networks (Hordijk and Steel 2004; Hordijk 2019) as a structural proxy of closure of the catalytic loop — counterfactual subtraction of individual components of the network is an operational realization of § 2.2 on the geochemical graph. (b) Isotopic signatures (fractionation of δ^{13}C, δ^{34}S) and chiral signatures — a proxy for C_v (the network's memory of the state of the environment). (c) An integrated thermodynamic-network test for positivity of self-payment via the measured free-energy balance of a RAF-closed subnetwork relative to its reactive surroundings. The development of such proxies and their calibration on contemporary biominiatures (minimal cells, synthetic protocells) is future work.

The connection with hydrothermal hypotheses is direct. These include the alkaline-vent scenarios of Martin and Russell (2003), the metabolic-core programs of Smith and Morowitz (2016), the dynamic kinetic stability of Pross (2012), and Kauffman's adjacent-possible (Kauffman 2019); the general physical framework of OoL is Walker (2017b). Each of these programs articulates the same structural transition from its own side, without a unifying thermodynamic measure. The constructive diagnostic is formulated as a direction of development, not as an already-available protocol. It requires the simultaneous appearance of three components: catalytic closure via RAF analysis of the geochemical network (Hordijk and Steel 2004; Hordijk 2019), informational memory via isotopic and chiral signatures, and a positive thermodynamic account by the scheme of § 5. This is an operational reformulation of the task of astrobiology: not only to "find where life is" but also to "detect the structural transition to it" — falsifiable in the event of discovering geochemical systems with a stable $\eta_v > 0$ without the characteristic triad, once the methodology is operationally further developed.

The ethics of artificial intelligence. The shift from the question "is the model complex enough to be a moral subject" to the question "will the model begin to bear its own errors itself." The appearance of one's own dissipative payment for the model, retained in the own degrees of freedom of the working loop, is the structural condition under which η_v emerges from the region of indeterminacy into positive values. The phasing pertains to the definition of the self-payment loop (the structural predicate of counterfactual ablation § 2.2 is binary), not to the empirical transition of the system; the empirical dynamics of the loop's appearance in a real population of AI systems is an open question with an expectedly continuous profile in the fraction of internalized payment. The current standard LLM architecture (a stateless forward pass with externalized payment of the infrastructure) excludes the fulfillment of the structural condition. The path to a positive η_v^{int} for an artificial system requires the simultaneous internalization of (a) a persistent state, (b) own payment of computation, (c) a physical error-return channel — not jointly realized in any existing system at the time of writing. This is a structural condition, not a calendar forecast. The formal anchoring to the triad. (a) A persistent state operationalizes $C_v^{\mathrm{int}} > 0$ (a memory channel for retention). (b) Own payment of computation — $T_v^{\mathrm{int}} > 0$ (one's own thermodynamic budget). (c) A physical error-return channel — $S_v^{\mathrm{int}} > 0$ (closure of the loop's topology). The simultaneity of all three yields $\eta_v^{\mathrm{int}} > 0$.

"To bear one's own errors" — in the technical sense of § 2.1: the model's errors return as a physical loss of free energy of the system itself, not as reputational sanctions through an external corporate substrate. A drop in the frequency of an LLM's usage and its retraining by the corporation's efforts is the corporation's payment, entering the η_v of the corporation; the η_v^{int} of the model as an agent remains zero. Regulator fines, legal sanctions against the developer, market reputational losses — return the damage not to the system retaining the model but to its external substrate.

Complex systems and social dynamics. In application to cities and civilizations, the scale opens two thresholds: the structural screening (I_v) and the verification (η_v). A civilization with high I_v and a sagging η_v — structurally rich, failing the retention of the loop — is diagnostically distinct from a civilization with a sagging screening (destruction of infrastructure). The differentiation is useful for studies of the resilience of social systems; it requires empirical verification.

The Fermi paradox and the externalization filter. The paper's hypothesis applied to the Fermi paradox: the silence of the Universe is explained not by the rarity of life but by **the collapse of technological civilizations at the threshold of externalization of the energy budget**. In the terms of the framework: a civilization passes the "filter" if and only if its η_v is positive through its own dissipation (T_v retained within its loop). Technological development directed toward the use of external sources (the power grid $\to$ reactors $\to$ megastructures) increases $E_{\text{actual}}^{\text{external}}$ without increasing $E_{\text{actual}}^{\text{own}}$ — most civilizations pass the I_v-screening (visible structural complexity: radio signals, megastructures) but **fail the η_v-verification** by the same mechanism as a stateless LLM under the boundary "weights at the moment of inference." Such civilizations are unstable on evolutionary timescales and do not persist as stationary SETI sources. The claim has the status of a speculative extrapolation and is subject to independent empirical verification. A testable statistical prediction about observable SETI signals: transient, non-repeating, a power-law distribution by duration with a cutoff — an indicator of the filter; a stationary spectrum with an IR excess, persisting over decade-long windows — an indicator of passing both thresholds. The existing null results of Dyson-sphere searches (the G-HAT survey over ~10^5 WISE galaxies, Griffith et al. 2015; the rationale for the program — Wright et al. 2014) are consistent with the filter prediction but do not refute alternative scenarios (rare-Earth-class hypotheses, technomaturity below the astrophysical resolution). The refutation condition: the discovery of three or more independent stationary SETI sources with a confirmed IR signature of megastructures over intervals $\gtrsim 50$ years is incompatible with the externalization-filter prediction in its current formulation.

The status of the consequence. The reformulation has the status of a speculative extrapolation: neither the NASA Astrobiology Strategy, nor the ESA Cosmic Vision, nor the planetary Decadal Survey operates with the category of "self-payment loop." If the standard programs achieve reliable demarcation of extraterrestrial life without recourse to η_v, the framework will bring no astrobiologically new discriminative power and reduces to the existing agnostic biosignatures.

Open questions. The non-stationary generalization of the stationary result on memory drift (§ 2.1) for systems with growing memory — biospheres over eonic windows, learning artificial systems — requires a rigorous proof that accounts for the cost of expanding memory (developed in the companion work, Andriishin 2026). The catalog of operational proxies for I_{pred} requires expansion with numerical examples and limits of applicability. The case of Gaia — the question of the unity of the biosphere's model — is open empirically and depends on data on the consistency of biogeochemical cycles over eonic windows. The operational protocol for detecting vital AI requires precise thresholds for population experiments. The philosophical status of η_v — a position of moderate operationalism with a demarcation between the ontological referent and the operational measure — is the task of a separate work.

Dormant states. Biological states of reversible arrest — spores, a tardigrade in cryo-anabiosis, frozen embryos, the lysogenic cycle of a phage — formally give $\dot{I}_{\text{pred}} = 0$ and $E_{\text{actual}} \to 0$, which takes η_v out of the domain of definition of the scale (§ 2.7). Reconciliation with biological convention ("spores are alive") is achieved through the averaging window τ_d: with a choice of τ_d on the evolutionary scale (the full cycle sporulation $\to$ activity $\to$ sporulation) the sign of η_v returns to positive. A formal extension of the scale to regimes of temporary reversibility of the sign is a direction of the companion work (Andriishin 2026).

Boundary conditions of applicability. The η_v scale does not make three types of claim.

1. It does not assert the identity of vitality and consciousness. High η_v is a necessary condition for complex cognitive activity (without retention of one's own predictive information, neither prediction, nor planning, nor reflection is possible), but not a sufficient one. The question of phenomenal experience, discussed within integrated information (Tononi et al. 2016) and the hard problem of consciousness, lies beyond the η_v scale.
2. It does not presuppose biocentrism by the reverse gesture. High η_v is possible in systems of a non-biological substrate — large cities, planetary-scale biospheres, hypothetical artificial embodied agents with a closed self-payment loop. This is a substantive prediction, not a calibration fit.
3. It does not reduce to a moral criterion. A civilization that fails the η_v-verification while retaining the I_v-screening is diagnosed as structurally rich but non-vital — a diagnosis of a structural state, not an ethical verdict. The ethical consequences of the program are a separate topic, resting on the structural diagnosis but not derivable from it by direct implication.

The stationary regime as an explicit limitation of the domain of applicability. All the results presented — definitions, demarcation tests on the paradigm cases, conditions of falsification — are formulated under the assumption of stationarity of the environment and finiteness of the system's memory. This is not a default simplification: stationarity stabilizes I_{mem} and renders the stepwise cost of retention (the Still bound (2)) stationary, and therefore permits sharply formulated demarcation answers. In the non-stationary regime — with growing memory, a drifting environment, and a nonempty archival layer — the formalism requires extension: a cumulative dissipative cost, accounting for the cost of retaining historical bits, and the phenomenon of informational nostalgia. This extension is the content of the companion work (Andriishin 2026). There a conditional lemma on the inevitability of nostalgia is proved in the canonical class of slow drift (OU-parameterization with FIFO-refresh and $c < 1$). The quantitative adiabatic asymptotics of $\dot{\eta_v}^{\text{excess}}$ for systems without periodic memory reset is formulated there as an open hypothesis, supported by a parametric scan. The informational capacity C_v of the present work is there methodologically refined through a partition of C_v^{static} (statistical complexity C_μ, structural memory — § 2.3) against $C_v^{\text{predictive}}$ (excess entropy $\mathbf{E}$, the predictive part — § 2.3), related by

$C_v^{\text{predictive}} = (1 - \nu) \cdot C_v^{\text{static}}$, where ν is the fraction of nostalgic bits; in the stationary regime of the present work the divergence of C_v^{static} and $C_v^{\text{predictive}}$ ceases to grow ($\dot{\nu} \to 0$, ν levels off at a constant value) and serves as a diagnostic of drift only in the non-stationary extension, which reconciles the two works. The applicability of the stationary η_v developed in the present article is limited to systems with an ergodic environment and finite memory; for biospheres over eonic windows, learning AI systems under drift, and civilizations with a growing archive — consult the non-stationary extension.

7 Conclusion

The work has defined vitality operationally as the pair (S, η_v): the predictive efficiency of self-modeling $\eta_v = I_{\text{pred}}/I_{\text{mem}} \in [0, 1]$ — the fraction of the memory retained by the system about the environment that still predicts its future — **given** a closed self-payment loop S (§ 2.7). The bound $\eta_v \leq 1$ for passive self-modeling is a consequence of the data processing inequality (without a separate thermodynamic premise; the active case — § 2.1), not an energetic normalization; the thermodynamic cost of retaining aging memory is borne by a separate anchor — the Still bound (2). Structural screening requires the simultaneous presence of three capacities — T_v, C_v, S_v — assembled into the composite index of structural readiness I_v. The index is a necessary but not sufficient condition of vitality; it occupies the same positional place as the assembly index, Φ, Deacon's morphodynamics. η_v itself is an operational criterion of synchronic vitality in the sense of a binary sign ($\eta_v > 0$ given a closed self-payment loop). The criterion distinguishes structurally rich but non-vital systems (the Sun, a hurricane, an LLM under the boundary "weights at the moment of inference" — η_v^{int} undefined) from systems with a structurally closed self-payment loop by the counterfactual ablation of § 2.2. Among systems with a closed loop, the bacterium has a reproducibly measured $\eta_v > 0$ via § 3.5. The city, the biosphere, the LLM-as-corporation — with a structurally positive sign ($S = 1$) at a quantitatively undefined magnitude of η_v (MI-estimation of I_{pred} and I_{mem} — an open problem, § 3.4, § 3.6, § 3.7). What is substantive across classes is precisely the sign, not the absolute magnitude (§ 3.5).

The application of the two-stage procedure to the six paradigm cases yields sharp demarcation answers and makes explicit the structural asymmetry: complexity is ubiquitous, closed self-payment loops are rare. The comparison with assembly theory, IIT, teleodynamics, and FEP positions η_v as an operational complement to each of the programs. Concrete conditions of falsification have been formulated, anchored to testable physical, informational, and evolutionary predictions.

The program is open to refutation along several independent lines, with the empirical risk concentrated in false-negative penetration and the carrying Prediction 1 (§ 5; § 2.7, the asymmetry of falsification; with a separately fixed condition confirming the uniqueness of the operationalization), and is disciplined against rescue maneuvers by the requirement to name the concrete failed capacity at each zeroing (§ 2.6): refutation of one capacity collapses both scales. This discipline limits but does not eliminate the Lakatosian risk of a protective belt (§ 2.6, § 2.7). In this lies its scientific status —

not a claim to a final definition of life, but a testable research program with explicitly articulated limits of applicability.

Biological consequences. The principal substantive effects of the η_v-program are within theoretical biology, not on the astrobiological or AI periphery. (a) The synchronic/diachronic decomposition of vitality (§ 2.7) gives a new formulation of the question "what is an organism" in the spirit of Pradeu (2018) and Bich and Green (2018): synchronic $\eta_v > 0$ is a necessary condition of the individuation of the living, while diachronic Darwinian dynamics is its complementary dimension. (b) Minimal cells (JCVI-syn3.0, Hutchison et al. 2016) and viruses receive an operationalizable boundary via the counterfactual ablation of § 2.2: the pair (WT *E. coli* / syn3.0) is distinguishable not by a qualitative criterion of organisational closure but by the quantitative scale η_v through both components (§ 4.7, § S4.7), while viral particles fall under the readdressing of the subject to (virion + host) — a diagnosis structurally homologous to the LLM-corp (B_4, § 3.4). (c) Major transitions in evolution (§ 4.8, § S4.8 — after the hypercycle and the major transitions of Maynard Smith–Szathmáry) are reformulated as events of readdressing the subject of self-payment with a thermodynamic criterion for detecting the event. A new aggregate actor appears, with its own $I_{\text{pred}}^{\text{new}}$, paid for by dissipation at the new level. (d) Symbiotic systems — plant–mycorrhiza (Simard et al. 1997; cf. the critical reassessment of Karst et al. 2023), coral–zooxanthellae — set up a test bench for the discipline of the conventionality of the boundary of § 2.2: whose η_v grows under symbiosis, how the sign depends on the choice of boundary between partners, under what conditions the symbiotic loop forms a new subject of self-payment. (e) Bioelectric morphogenesis (Levin 2019) — bioelectric patterns as M_X (the physical realization of the model of morphology), paid for by ionic currents through cell membranes — gives a direct candidate for η_v-measurement in the morphogenetic field. The power-law dependence of the rate of evolutionary adaptation on η_v (Prediction 1, § 5) is testable on a single microbial line in ordinary experimental evolution, without requiring an astrobiological or AI context, and constitutes the program's central empirical prediction for theoretical biology.

Speculative consequences (astrobiology, AI ethics, demarcation). Beyond the biological core, the program offers a reformulation of traditionally metaphysical questions. The search for extraterrestrial life acquires a thermodynamic requirement of closed self-payment loops on top of the existing chemical biosignatures (a complement to the Ladder of Life Detection, not a replacement). The ethical status of artificial systems is posed as a question about the appearance of one's own dissipative payment for the model (alongside the traditional lines of sentience / welfare / functionalism). The boundary between the structurally complex and the vital receives an operational *necessary* cross-section: $\eta_v > 0$ is a necessary condition of synchronic vitality (§ 2.7); the upper bound $\eta_v \leq 1$ is an **unconditional theorem** (a consequence of the data processing inequality, § 2.1) for passive self-modeling; the active case (sensorimotor feedback) is an open problem (directed information, the Sagawa–Ueda 2010 bound). A full definition of life in the sense of the NASA formula (Joyce 1994) also requires Darwinian dynamics at the population level, and both dimensions — synchronic and diachronic — are complementary. Each of these reformulations is not an automatic solution of the corresponding question; it is a pointer to a concrete measurable quantity relative to which the question can be formulated in a disciplined way.

For theoretical biology, η_v opens a quantitative bridge between the tradition of organisational closure (autopoiesis, M,R-systems) and predictive models of adaptation.

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Data Availability. No primary empirical data were generated for this work. The reproducible code for the worked examples (§§ 3.4–3.7, 4.4: *E. coli* chemotaxis, the bimetallic thermostat, synthetic city profiles, the biosphere-as-compressor, and the LLM-as-corporation) is openly available in the GitHub repository `andriishin/landauer-self-modeling-oa` (https://github.com/andriishin/landauer-self-modeling-oa) and, together with the preprint of this manuscript, is archived on Zenodo (DOI: 10.5281/zenodo.21039160).

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